DocLang: Universal AI Document Format

Version: 0.9

Contents

Introduction 5
Scope 6
Normative references 7
Terms and definitions 8
  Abstract concepts 8
  Roles 8
  Adopted from XML 8
Symbols and abbreviated terms 9
Conformance 10
  Version 10
  Omitted version 10
  Compatibility 10
  Schema versions 10
Language design principles 11
  Property semantics 11
  Head and body areas 11
    Element head 11
    Element body 12
    Document head 12
    Document body 12
    Head and body example 12
  Content encoding and whitespace handling 12
  Subclasses 14
Language specification 15
Usage examples 16
  Simple document structure 16
  Pictures 16
  Code snippets 17
  Math and equations 18
  Lists 20

Tables ... 22
Fields ... 24
Field structure rules ... 24
Field region examples ... 25
Tracks ... 44
Chapters ... 46
Split structure ... 47
Cross-column structure example ... 47
Cross-page structure example ... 48
References ... 50
Cross-references ... 50
Hyperlinks ... 50
Formatting ... 51
Page break with continuation ... 52
Custom vocabularies ... 53
Annexes ... 54
Annex A (normative): Reference ... 54
Special elements ... 54
Semantic elements ... 55
Property elements ... 66
Payload elements ... 70
Formatting elements ... 73
Structural elements ... 75
Document head elements ... 79
Annex B (normative): DocLang archive format ... 79
Layout ... 79
Page alignment ... 81
Track alignment ... 81
Annex C (informative): Recommendations ... 81
Recommended labels ... 81
Custom vocabulary naming and namespacing ... 82
Token vocabulary ... 82
Annex D (informative): Future extensions ... 86

Horizontal threading 86
Explicitly wrapped repetitive items 86
Multi-part track media 86
Metadata 86

Introduction

Digital documents exist in many formats, which makes processing, conversion, and understanding challenging. Moreover, many of these formats have been designed primarily for rendering, so semantic information, structural relationships, and geometric context may be lost during document conversion.

DocLang aims to address these challenges by providing a minimalist, unambiguous markup format that:

- Preserves complete document structure and semantics
- Maintains geometric and layout information when appropriate
- Supports complex document components including tables, formulas, code, nested lists, and charts
- Enables lossless round-trip conversion between formats regarding content
- Maintains token efficiency by defining controlled vocabulary of tags and attributes
- Eliminates ambiguity by enforcing a well defined set of tags with restricted, non generic usage (and preserves semantic clarity)

This document builds upon research in document understanding and is intended to represent the content of a document as accurately as possible while maintaining implementation simplicity.

Scope

This document specifies DocLang, a markup language for representing structured document content, including its semantic, geometric, and formatting information.

It covers the following aspects:

- The syntax and semantics of the DocLang markup language
- Rules for encoding document structure, content, and metadata
- Primitives for representing geometric layout and pagination
- Methods for expressing formatting and text direction
- Specifications for complex document components (tables, charts, formulas, code, forms)

Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

1. Optimized Table Tokenization for Table Structure Recognition (OTSL), https://doi.org/10.1007/978-3-031-41679-8_3
2. W3C XML 1.0 Specification (Fifth Edition)
3. W3C XML Schema Definition Language (XSD) 1.0 (Second Edition): Part 1 (Structures) and Part 2 (Datatypes)
4. IETF RFC 3629: UTF-8, a transformation format of ISO 10646
5. IETF RFC 3986: Uniform Resource Identifier (URI): Generic Syntax
6. IETF RFC 2397: The "data" URL scheme
7. Semantic Versioning 2.0.0 (semver.org)

Terms and definitions

For the purposes of this document, the following terms and definitions apply. Terms adopted from XML are as defined in [2].

Abstract concepts

8. **DocLang instance**

Document content (e.g. text, tables, images) expressed in DocLang markup.

9. **document component**

A cohesive and meaningful part of document content, e.g. a table, list item with a marker, a bold piece of text, etc.

Roles

An implementation may act in one or more of the following roles.

10. **producer**

An implementation that creates DocLang instances.

11. **processor**

An implementation that reads, transforms or renders DocLang instances.

12. **validator**

A processor that checks DocLang instances against this document.

Adopted from XML

13. **element**

An XML element.

14. **attribute**

An XML attribute.

Note 1 to entry: When referring to an attribute in prose, this document uses XPath-style notation: `element@attribute` (e.g. `label@value`, `location@resolution`).

15. **tag**

An XML tag: can be a start-tag, an end-tag, or an empty-element tag (a.k.a. self-closing tag).

Symbols and abbreviated terms

CDATA Character data

LLM Large language model

OTSL Optimised table-structure language

URI Uniform resource identifier

URL Uniform resource locator

UTF Unicode transformation format

XML Extensible Markup Language

XSD XML Schema Definition

Conformance

A DocLang instance conforms to this document if it satisfies all the requirements specified in it. Schemas for checking instances (an XSD [3] schema and Schematron rules) are provided with the reference toolkit of the DocLang Project. They are aids, not a definition of conformance: in case of discrepancies, this document is authoritative.

Requirements addressed to a producer, processor or validator apply to implementations acting in that role. A producer SHOULD ensure that the instances it produces conform, for example by validating its output and rejecting or repairing non-conforming results.

Version

A DocLang instance states the version of this document it is intended to conform to, in `MAJOR.MINOR` format, through the `version` attribute of the root `<doclang>` element.

Omitted version

The `version` attribute is optional. If it is omitted, the version of the instance is unspecified: it is neither implied to be the version of this document nor any earlier version.

- A producer SHOULD always set `version`.
- A validator SHOULD validate an instance with an unspecified version against the latest version of this document it supports, and SHOULD warn that no version was declared.
- A validator MAY allow the user to explicitly select the version to validate against, overriding the declared one.

Compatibility

The versions of this document follow the principles of Semantic Versioning [7] (considering X >= 1, and Y < Z):

- A DocLang instance that conforms to version `X.Y` also conforms to version `X.Z`.
- A DocLang instance that conforms to version `X.Z` need not conform to version `X.Y`.

Example:

- A `1.0` instance conforms to version `1.1`
- A `1.1` instance need not conform to version `1.0`

Schema versions

A schema may additionally carry a patch version, i.e. a full Semantic Versioning string (e.g., `1.0.0`, `1.0.1`), to track schema-level changes that do not affect document compatibility.

The XSD schema constrains the `version` attribute of the root `<doclang>` element only to the `MAJOR.MINOR` format; selecting the schema that corresponds to an instance's version is the responsibility of the validator.

Language design principles

Property semantics

In XML, the attribute syntax allows explicitly separating an element's properties from its content. Let's consider the following example: `<elem size="250" color="#ffeedd">foo</elem>`. This denotes an `elem` element, including its properties (`size` and `color`) and its content (`foo`).

For an element with multiple possible property values, the attribute syntax can lead to an increased complexity of the respective possible tokenized representations. For instance, the following could all be valid variants of an `elem` start tag: `<elem size="300" color="#aabbcc">`, `<elem size="42">`, `<elem color="#112233">`, `<elem>`.

Aiming at LLM-friendliness, in such cases, DocLang often favors an alternative representation of property semantics, namely captured as respective elements leading the content. The example above could be represented as `<elem><size>250</size><color>#ffeedd</color>foo</elem>`. Depending on the specific properties, empty elements are used too.

This representation can reduce the number of tokens and streamline how XML is mapped to them, making it easier for language models to learn and predict.

For elements with a strictly limited set of possible property values, attributes are still used.

Head and body areas

Documents and their core components (as implemented via semantic elements) can have various properties associated with them. To separate between properties and actual content, DocLang follows a two-part scheme, both on the the component / element level and on the global document level.

Element head

The XML content of a semantic element begins with an *element head*, which is a sequence of dedicated elements that establish the element's properties, namely in this order:

- `<label>` (optional)
- `<thread>` (optional)
- `<xref>` or `<href>` (mutually exclusive, optional)
- `<layer>` (optional)
- optional sequence of 4 `<location>`s, whereby values are interpreted in alternating axis order, as `x_min, y_min, x_max, y_max` (after resolution normalization), w.r.t. the top-left corner of the page
- `<caption>` (optional, may be repeated)
- `<description>` (optional)

- `<summary>` (optional)
- `<custom>` (optional)

Element body

The XML content after the element head is called the *element body* and contains the effective payload of the semantic element.

Document head

Root element `<doclang>` begins with an optional *document head*, which encapsulates any global document properties in a dedicated `<head>` element.

Document body

The remaining XML content after the optional document head is called the *document body* and contains the effective payload of the DocLang instance.

Head and body example

While the details are specified in the sections further below, this snippet shows an example of this scheme:

```
<doclang>
  <!-- document head: -->
  <head>
    <default_resolution width="1024" height="1024"/>
    <!-- ... -->
  </head>

  <!-- document body: -->
  <text>

    <!-- element head: -->
    <location value="60"/><location value="260"/>
    <location value="440"/><location value="270"/>

    <!-- element body: -->
    Hello world!

  </text>
  <text>
    <!-- element body: -->
    Headless paragraph
  </text>
</doclang>
```

Content encoding and whitespace handling

As DocLang is XML, standard XML [2] encoding rules apply — for example:

- any provided XML prolog defines the encoding, otherwise UTF-8 [4] is assumed

- special characters reserved by XML, such as `<`, can be represented either by escaping with the respective XML entities (e.g. `<` becomes `<`) or by using CDATA section syntax (e.g. raw text `<foo>` can be represented as `<![CDATA[<foo>]]>`.)

In this section, *whitespace* means any of the following characters:

- space (U+0020)
- tab (U+0009)
- carriage return (U+000D)
- line feed (U+000A)

The rules below do not apply to attribute values, which are subject to XML attribute-value normalization.

`<content>` preserves whitespace: its content shall be interpreted literally.

A *text block* is defined as any of the following:

- the body of an element in which raw text is allowed directly (raw text content type "Allowed" in Annex A), unless that element is part of another text block (e.g. `<bold>` within `<text>`);
- the body of a virtual `<text>`;
- the spoken text of a transcript turn (see `<track>`).

When reading a text block, XML comments and processing instructions within it, as well as the element heads of the elements within it, are disregarded, as if they were removed. The elements of such element heads are read separately: e.g. a `<caption>` forms a text block of its own, to which the same rules apply.

A consumer shall interpret whitespace outside `<content>` as follows:

16. Whitespace outside text blocks shall be ignored.
17. Whitespace at the start and at the end of a text block shall be ignored.
18. Any other sequence of consecutive whitespace, taken as a whole, shall be interpreted as a single space. Whitespace separated only by start-tags and end-tags of elements within the text block that allow raw text (e.g. formatting elements) counts as consecutive; whitespace within a `<content>` element does not count together with whitespace outside it.
19. When a sequence of whitespace spans several elements, the space shall be attributed to the element that contains the first character of the sequence.

To retain whitespace that these rules would collapse or ignore, such as runs of spaces or line breaks, a producer shall use `<content>`. A producer should use `<content>` only where these rules would otherwise change the intended reading. Conversely, a producer can insert a line break and indentation wherever a single space is intended, for example to improve readability.

Example:

```
<text>
  Water (H<subscript>2</subscript>O) is <bold>clear </bold>
  <italic> and odorless</italic>.
  This keeps <content>two  spaces</content> as written.
</text>
```

This reads as `Water (H₂O) is clear and odorless. This keeps two  spaces as written.`, with:

Part	Reason
no space before `Water`, none after `written.`	rule 2 (start and end of the text block)
`H₂O` stays tight	no whitespace in the source, so none in the reading
one space between `clear` and `and`, which is bold	rules 3 and 4: one sequence across the tags; its first character is in `<bold>`
one space between `odorless.` and `This`	rule 3: the line break and indentation read as a single space
`two  spaces` keeps both spaces	`<content>` is literal

Subclasses

For core document components like `<picture>` or `<code>`, a subclass may be indicated via the `<label>` element. DocLang provides some recommended value domains (see Recommendations), but for extensibility purposes, the label value is not to be validated. Additionally, some elements, e.g. `<picture>`, may include a `class` attribute for providing an intermediate classification level typically associated with specific semantics and structural implications.

In the example further below:

- `class="chart"` means the picture is semantically considered a chart and is therefore associated with special structuring rules (in the case of a chart, it can namely contain structured chart data in OTSL format)
- the label value is conveying the specific chart subclass for further classification purposes

```
<doclang>
  <picture class="chart">
    <label value="bar_chart"/>
    <tabular><!-- structured chart data ... --></tabular>
  </picture>
</doclang>
```

Language specification

DocLang markup is encoded as XML. Recommended file extension: **`.dclg`**.

The individual DocLang elements and attributes, as well as DocLang's contextual rules are specified in Reference.

The DocLang archive format is defined in DocLang Archive Format.

Non-normative recommendation guidelines are covered in Recommendations.

Planned extensions are discussed in Future extensions.

A reference toolkit for DocLang is provided by the DocLang Project.

Usage examples

Simple document structure

In the simplest document example, document elements are in a flat list,

```
<doclang>
  <heading>Research Paper Title</heading>
  <heading level="2">Abstract</heading>
  <text>This paper presents...</text>
  <heading level="2">Introduction</heading>
  <text>In recent years...</text>
  <heading level="3">Background</heading>
  <text>Previous work has shown...</text>
</doclang>
```

In case of page-layout information, the coordinates are provided only at the semantic element level. Coordinates are not allowed at the group level.

```
<doclang>
  <heading>
    <location value="10"/><location value="20"/><location value="30"/><location
value="40"/>
    Research Paper Title
  </heading>

  <heading level="2">
    <location value="10"/><location value="20"/><location value="30"/><location
value="40"/>
    Abstract
  </heading>
  <text>
    <location value="10"/><location value="20"/><location value="30"/><location
value="40"/>
    This paper presents...
  </text>

  <heading level="2">Introduction</heading>
  <text>In recent years...</text>

  <heading level="3">Background</heading>
  <text>Previous work has shown...</text>
</doclang>
```

Pictures

Pictures are captured with the `<picture>` element and may be specialized via the `class` attribute, as detailed in the reference. The picture data can be provided via the `<src>` element as a URI [5] capturing the image either by reference (e.g. https URL) or as base64-encoded data (data URI [6]).

Picture by reference URI:

```
<picture>
  <src uri="https://example.com/image.jpg"/>
</picture>
```

Picture by base64-encoded data:

```
<picture>
  <src
uri="
AQ0AAADCoPdPbQ8HFAAAAAAAAAAAAAAA8G4wQAABiwCo9wAAAABJRU5ErkJggg==">
</picture>
```

Bar chart using recommended label and `<tabular>` for capturing structured chart data:

```
<picture class="chart">
  <label value="bar_chart"/>
  <src uri="chart.svg"/>
  <tabular>
    <ched/>Category<ched/>Value<nl/>
    <fcel/>A<fcel/>10<nl/>
    <fcel/>B<fcel/>20<nl/>
  </tabular>
</picture>
```

Picture with document caption, description, and summary:

```
<picture>
  <caption>FIG. 2. System architecture</caption>
  <description>
    A block diagram with a browser client, an application server, and a database.
    Arrows show HTTP requests from client to server and SQL queries from server to
database.
  </description>
  <summary>The system uses a three-tier architecture.</summary>
  <src uri="fig2.png"/>
</picture>
```

Code snippets

Code content is captured with `<code>`, either as a standalone block or inlined within a semantic element. For language classification, use a `<label>` in the element head (see Recommendations).

Whitespace can be retained by `<content>` and XML escape characters can be addressed using CDATA.

Inline code:

```
<text>
  Check your version with <code>python --version</code>.
  We can even have <code><bold>formatted code</bold></code>.
</text>
```

Code block with language classification and whitespace preservation:

```
<code>
  <label value="Python"/>
  <content>def add(a, b):
    return a + b

if __name__ == "__main__":
    print(add(2, 3))</content>
</code>
```

Grouped code with caption and coordinates:

```
<code>
  <label value="JavaScript"/>
  <location value="10"/><location value="80"/><location value="400"/><location
value="300"/>
  <caption>
    <location value="10"/><location value="20"/><location value="400"/><location
value="60"/>
    Listing 1: Minimal HTTP server
  </caption>
  <content><![CDATA[// Minimal Node.js server
import http from 'node:http';
const server = http.createServer((req, res) => {
  res.end('OK');
});
server.listen(3000);]]></content>
</code>
```

Long code blocks can be split across pages using continuation elements; keep `<label>` in the first fragment.

```
<code>
  <label value="Shell"/>
  <thread thread_id="42"/>
  <content># Part 1
seq 1 5 | while read n; do echo $n; done</content>
</code>
<page_footer>Foo</page_footer>
<page_break/>
<code>
  <thread thread_id="42"/>
  <content>echo "done"</content>
</code>
```

Math and equations

All math is authored as LaTeX inside <formula>, whether in standalone blocks or inlined within a semantic element (similar to code).

Inline formula:

```
<text>
  The famous relation <formula>E = mc^2</formula> connects mass and energy.
  For small x, <formula>\sin x \approx x - x^3/3!</formula> holds.
  The binomial: <formula>(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k</formula>.
</text>
```

Block Formula:

```
<formula>
  \int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}
</formula>
```

Formula with caption and coordinates:

```
<formula>
  <location value="10"/><location value="80"/><location value="400"/><location
value="150"/>
  <caption>
    <location value="10"/><location value="20"/><location value="400"/><location
value="60"/>
    Equation for the normal distribution
  </caption>
  f(x) = \frac{1}{\sigma\sqrt{2\pi}}\,\exp\!\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)
</formula>
```

Multi-line LaTeX in a single formula:

```
<formula>
  <content><![CDATA[\begin{align}
  \nabla\cdot\mathbf{E} &= \frac{\rho}{\varepsilon_0} \\
  \nabla\cdot\mathbf{B} &= 0
\end{align}]]></content>
</formula>
```

Cross-page block formula with continuation:

```
<formula>
  <thread thread_id="42"/>
  <content>\begin{equation}
  \mathbf{F}(t) = \int_0^t e^{A(t-\tau)}\,\mathbf{b}(\tau)\,d\tau
\end{equation}</content>
</formula>
<page_footer>Foo</page_footer>
<page_break/>
<formula>
  <thread thread_id="42"/>
  <content>% Optional tail content if the printed equation spans pages</content>
</formula>
```

Note: All math content is LaTeX; omit `$...$` or `\[...\]` delimiters since the tag conveys math context.

Lists

A list can contain as list items any semantic element sequence. List items are namely introduced by `<ldiv>` elements, whereby an `<ldiv>` may optionally contain a marker. This means that a non-empty `<list>` element body must begin with an `<ldiv>`.

To promote token efficiency within the repetitive context of a list, a list item (the content between two sibling `<ldiv>`s or until `</list>`) may also be a virtual `<text>`, i.e. unwrapped text content.

Basic list with virtual `<text>`

```
<list>
  <ldiv/>First item
  <ldiv/>Second item
  <ldiv/>Third item
</list>
```

Unordered list with optional markers

```
<list>
  <ldiv><marker>•</marker></ldiv>
  <text>First item with <bold>bold</bold> text</text>
  <ldiv>
    <!-- Marker with its own coordinates -->
    <marker>
      <location value="50"/><location value="110"/><location value="60"/><location
value="120"/>
      •
    </marker>
  </ldiv>
  <text>Second item</text>
</list>
```

Ordered list; markers are optional and can hold the printed numbering

```
<list class="ordered">
  <ldiv><marker>1.</marker></ldiv>
  <text>Install dependencies</text>
  <ldiv><marker>2.</marker></ldiv>
  <text>Run tests</text>
  <ldiv/>
  <!-- No marker provided -->
  <text>Ship release</text>
</list>
```

Checkbox items with selection state; markers optional

```
<list>
  <ldiv><marker><checkbox class="selected"/></marker></ldiv>
  <text>Completed task</text>
  <ldiv><marker><checkbox class="unselected"/></marker></ldiv>
```

```
  <text>Pending task</text>
</list>
```

Nested lists (mixing ordered and unordered)

```
<list class="ordered">
  <ldiv><marker>1.</marker></ldiv>
  <text>Setup project</text>
  <list>
    <ldiv><marker>•</marker></ldiv>
    <text>Create virtual environment</text>
    <ldiv><marker>•</marker></ldiv>
    <text>Configure linter</text>
  </list>
  <ldiv><marker>2.</marker></ldiv>
  <text>Implement features</text>
</list>
```

Page breaks and continuation

Lists can span multiple pages. Use `<thread thread_id="..."/>` to indicate continuation. You may thread the whole list and, if a particular component, e.g. a `text` is broken, also thread the component itself.

List split across pages

```
<list class="ordered">
  <thread thread_id="1"/>
  <ldiv><marker>1.</marker></ldiv>
  <text>First item</text>
  <ldiv><marker>2.</marker></ldiv>
  <text>Second item</text>
</list>
<page_break/>
<list class="ordered">
  <thread thread_id="1"/>
  <ldiv><marker>3.</marker></ldiv>
  <text>Third item</text>
</list>
```

Single list-item broken by a page break

```
<list>
  <thread thread_id="2"/>
  <ldiv><marker>•</marker></ldiv>
  <text>
    <thread thread_id="3"/>
    This item starts on page 1 and continues
  </text>
</list>
<page_break/>
<list>
  <thread thread_id="2"/>
```

```
  <ldiv/>
  <text>
    <thread thread_id="3"/>
    on page 2 until it ends.
  </text>
</list>
```

Notes

- `<marker>` is optional. Include it when the printed glyph/number is visible.
- `<marker>` may begin with an element head (e.g. `<location>` coordinates to pinpoint bullet/number placement) before the visible glyph or `<checkbox>`.
- Like a `<text>`, a `<marker>` may also hold semantic elements, e.g. a `<picture>` for an image bullet.
- Lists can nest as shown above.
- When broken across pages, close items before the `page_break`, then re-open and continue with matching `thread` ids after the break.

Tables

A table is defined by a `<table>` element, that contains cells, as delimited by the respective OTSL [1] structural elements (e.g., `<fcel/>`, `<ched/>`). This means that a non-empty `<table>` element body must begin with such an OTSL structural element.

Similarly to lists above, while a cell can naturally contain any semantic element sequence, it may also be a virtual `<text>`, i.e. unwrapped text content.

Basic table with virtual `<text>`:

```
<table>
  <ched/>Method<ched/>Accuracy<nl/>
  <fcel/>Baseline<fcel/>0.8<nl/>
  <fcel/>Proposed<fcel/>0.92<nl/>
</table>
```

Example with table location and semantic elements:

```
<table>
  <location value="40"/><location value="130"/><location value="400"/><location
value="320"/>
  <ched/>Method<ched/>Accuracy<nl/>
  <fcel/>Baseline<fcel/>0.85<nl/>
  <fcel/>Proposed<fcel/>0.92<nl/>
</table>
```

Example with caption:

```
<table>
  <location value="40"/><location value="130"/><location value="400"/><location
value="320"/>
  <caption>
```

```
    <location value="40"/><location value="80"/><location value="400"/><location
value="110"/>
    Table 1: Experimental Results
  </caption>
  <ched/>Method<ched/>Accuracy<nl/>
  <fcel/>Baseline<fcel/>0.85<nl/>
  <fcel/>Proposed<fcel/>0.92<nl/>
</table>
```

Immediately after a cell-creating element (e.g., `<fcel/>`, `<ched/>`), place the cell's content, which may include any semantic element, such as `text`, `list`, even nested `table` or `picture` elements.

```
<table>
  <location value="40"/><location value="200"/><location value="400"/><location
value="400"/>
  <caption>
    <location value="40"/><location value="140"/><location value="400"/><location
value="180"/>
    Table 3: Rich Cells
  </caption>
  <ched/><text>Description</text><ched/><text>Details</text><nl/>
  <fcel/>
  <text>Pipeline steps</text>
  <fcel/>
  <list>
    <ldiv/>
    Ingest
    <ldiv/>
    Process
    <ldiv/>
    Export
  </list>
  <nl/>
  <fcel/>
  <text>Nested table</text>
  <fcel/>
  <table>
    <caption>Inner table</caption>
    <ched/>
    <text>Key</text>
    <ched/>
    <text>Value</text>
    <nl/>
    <fcel/>
    <text>A</text>
    <fcel/>
    <text>1</text>
    <nl/>
    <fcel/>
    <text>B</text>
    <fcel/>
    <text>2</text>
```

```
    <nl/>
  </table>
  <nl/>
  <fcel/>
  <text>Image</text>
  <fcel/>
  <picture>
    <caption>Example image inside a cell</caption>
    <src uri="assets/img/sample.png"/>
  </picture>
  <nl/>
</table>
```

Notes:

- OTSL follows the rectangular rule; ensure each row has the same number of structural elements up to `<nl/>`.
- A cell can include any valid DocLang semantic element sequence.

Fields

Fields provide a flexible structure for representing key-value data and structured content. The field elements allow for more flexible document structures where keys and values may be separated by other content or organized in complex layouts.

element	Description
`<field_region>`	Field region container; can contain `field_item` elements as descendants
`<field_item>`	Field item container; may contain 0-1 `key` and 0-many `value` elements as descendants
`<field_heading>`	Field heading within a field region; has an optional attribute `level`
`<key>`	Key of the field item: can be a descendant of `field_item`
`<value>`	Value of the field item: can be a descendant of `field_item`; optional `class` attribute with values `read_only` (default) or `fillable`
`<hint>`	Hint for a fillable value field; recommended to be used within the context of a `field_item`; can describe a format, example, or additional description

Note that all of the above are semantic elements and can therefore contain their own element head, including e.g. their own bounding box location information.

Field structure rules

- Any `field_heading` or `field_item` must be a descendant of a `field_region` (not necessarily a direct child)

- Any `key` or `value` element must be a descendant of a `field_item` (not necessarily a direct child)

— More precisely, 0 or 1 `key` element and 0 or more `value` elements are allowed within a single `field_item`

    - For a given `field_item`, the 0..1 `key` constraint applies to its own descendant scope, excluding descendants that belong to nested `field_item` elements
  - This way, the `field_item` can serve for associating the `values` with a `key`
- The `value` element has an optional `class` attribute:
  - `class="read_only"` (default): Indicates a pre-filled, non-editable value
  - `class="fillable"`: Indicates an empty or editable field that can be filled in
- The `hint` element is recommended to be used within a `field_item` context to provide guidance for relevant field elements, such as fillable values.
- All field elements are semantic elements and can therefore contain their own element head, including e.g. their own bounding box location information.

Field region examples

Basic field region with simple key-value pairs:

```
<field_region>
  <field_item>
    <key>Name:</key>
    <value>John Smith</value>
  </field_item>
  <field_item>
    <key>Date of Birth:</key>
    <value>1985-03-15</value>
  </field_item>
  <field_item>
    <key>Address:</key>
    <value>123 Main Street, Anytown, USA</value>
  </field_item>
</field_region>
```

Field region with headings and complex layout:

```
<field_region>
  <field_heading>Personal Information</field_heading>

  <field_item>
    <text><key>Full Name:</key></text>
    <text><value>Jane Doe</value></text>
  </field_item>

  <field_item>
    <text><key>Email:</key></text>
    <text><value>jane.doe@example.com</value></text>
  </field_item>
```

```
  <field_heading level="2">Employment Details</field_heading>

  <field_item>
    <text><key>Company:</key></text>
    <text><value>Acme Corporation</value></text>
  </field_item>

  <field_item>
    <text><key>Position:</key></text>
    <text><value>Senior Engineer</value></text>
  </field_item>
</field_region>
```

Field item with multiple values:

```
<field_region>
  <field_item>
    <key>Phone Numbers:</key>
    <value>+1-555-0100 (Home)</value>
    <value>+1-555-0101 (Work)</value>
    <value>+1-555-0102 (Mobile)</value>
  </field_item>
</field_region>
```

Field item without a key (value-only):

```
<field_region>
  <field_heading>Additional Notes</field_heading>
  <field_item>
    <value>This is a standalone value without an explicit key.</value>
    <value>Multiple values can be provided.</value>
  </field_item>
</field_region>
```

Fillable fields with hints:

```
<field_region>
  <field_heading>Application Form</field_heading>

  <field_item>
    <key>Full Name:</key>
    <value class="fillable"></value>
    <hint>Enter your first and last name</hint>
  </field_item>

  <field_item>
    <key>Date of Birth:</key>
    <value class="fillable"></value>
    <hint>Format: YYYY-MM-DD</hint>
  </field_item>

  <field_item>
    <key>Email Address:</key>
```

```
    <value class="fillable"></value>
    <hint>example@domain.com</hint>
  </field_item>

  <field_item>
    <key>Status:</key>
    <value class="read_only">Pending Review</value>
  </field_item>
</field_region>
```

Field region with mixed content:

```
<field_region>
  <location value="50"/><location value="100"/>
  <location value="500"/><location value="400"/>

  <field_heading>Product Specifications</field_heading>

  <text>The following specifications apply to Model XYZ-2000:</text>

  <field_item>
    <key>Dimensions:</key>
    <value>10cm × 15cm × 5cm</value>
  </field_item>

  <field_item>
    <key>Weight:</key>
    <value>250g</value>
  </field_item>

  <picture>
    <caption>Product diagram showing key components</caption>
    <src uri="assets/product-diagram.png"/>
  </picture>

  <field_item>
    <key>Materials:</key>
    <list>
      <ldiv><marker>•</marker></ldiv>
      <text><value>Aluminum alloy frame</value></text>
      <ldiv><marker>•</marker></ldiv>
      <text><value>Tempered glass display</value></text>
      <ldiv><marker>•</marker></ldiv>
      <text><value>Silicone rubber grips</value></text>
    </list>
  </field_item>
</field_region>
```

Simple key-values

Firma:	***Holcim Kies und Splitt GmbH***	Datum:	**23.08.2019**	Blatt- Nr.:	**1 / 2**
Werk:	**Schalkholz**	Werks- Nr.:	**303.61 K**	Petrograph. Typ:	**Quartiärer Sand + Kies**

```
<field_region>
    <field_item>
        <key>Firma:</key>
        <value>Holcim ... GmbH</value>
    </field_item>
    <field_item>
        <key>Datum:</key>
      <value>23.08.2019</value>
    </field_item>
    <!--...-->
    <field_item>
        <key>Petrograph. Typ:</key>
        <value>Quartiarer Sand + Kies</value>
    </field_item>
</field_region>
```

Nesting forms and using form headings

14. Transport information

Land transport ADR/RID and GGVS/GGVE (Germany)
GGVS/GGVE class: 8 Number and letter: 1B
ADR/RID class: 8 Number and letter: 1B
Name of material: 2796 SCHWEFELSAEURE

River transport ADN/ADNR
not examined

Sea transport IMDG
IMDG class: 9 UN-No.: 3316 Packaging group: II
Ems: 8-06 MFAG: 700
Correct technical name: CHEMICAL KIT

Air transport ICAO-TI and IATA-DGR
ICAO/IATA class: 9 UN/ID-No.: 3316 Packaging group: II
Correct technical name: CHEMICAL KIT

The transport regulations are cited according to international regulations and in the form applicable in Germany (GGVS/GGVE). Possible national deviations in other countries are not considered.

THESE TRANSPORT DATA APPLY TO THE ENTIRE PACK !

```
<field_region>
  <field_heading>
      <marker>14.</marker>
      Transport Information
  </field_heading>
  <field_region>
      <field_heading>
          Land transport ... (Germany)
      </field_heading>
      <field_item>
          <key>GGVS/GGVE class:</key>
          <value>8</value>
      </field_item>
      <field_item>
          <key>ADR/RID class:</key>
          <value>8</value>
      </field_item>
```

```
      <!--...-->
  </field_region>
  <field_item>
      <key>River transport ADN/ADNR</key>
      <value>not examined</value>
  </field_item>
  <field_region>
      <field_heading>
          Sea transport IMDG
      </field_heading>
      <!--...-->
  </field_region>
  <!--...-->
  <text>
      The transport ... considered.
  </text>
  <text>
      THESE TRANSPORT ... PACK!
  </text>
</field_region>
```

Fillable form

<table><tr><td>

```
<field_region>
    <field_item>
        <key>Description</key>
        <value>A.A. Cat</value>
    </field_item>
    <field_item>
        <key>Quant.</key>
        <value></value>
    </field_item>
    <field_item>
        <key>Un</key>
        <value></value>
    </field_item>
    <field_item>
        <key>Measure</key>
        <value></value>
    </field_item>
    <field_item>
        <key>Price (in currency)</key>
        <value></value>
    </field_item>
    <field_item>
        <key>Un</key>
        <value></value>
    </field_item>
    <field_item>
        <key>Total</key>
        <value></value>
    </field_item>
```

```
    <text></text>
    <field_region>
        <field_item>
            <key>Delivery Cost</key>
            <value></value>
        </field_item>
        <field_item>
            <key>Maintenance</key>
            <value></value>
        </field_item>
        <!--...-->
    </field_region>
    <field_region>
        <field_item>
            <key>Date and time of delivery:</key>
            <value></value>
        </field_item>
        <!--...-->
        <field_item>
            <key>Guarantee</key>
            <value></value>
        </field_item>
        <text>
            Delivery Suppl...Finance Department
        </text>
    </field_region>
    <!--...-->
</field_region>
```

</td><td style="vertical-align: top;">

Description	A.A.Cat.		Quant.	Un Measure	Price (in currency) Un	Total

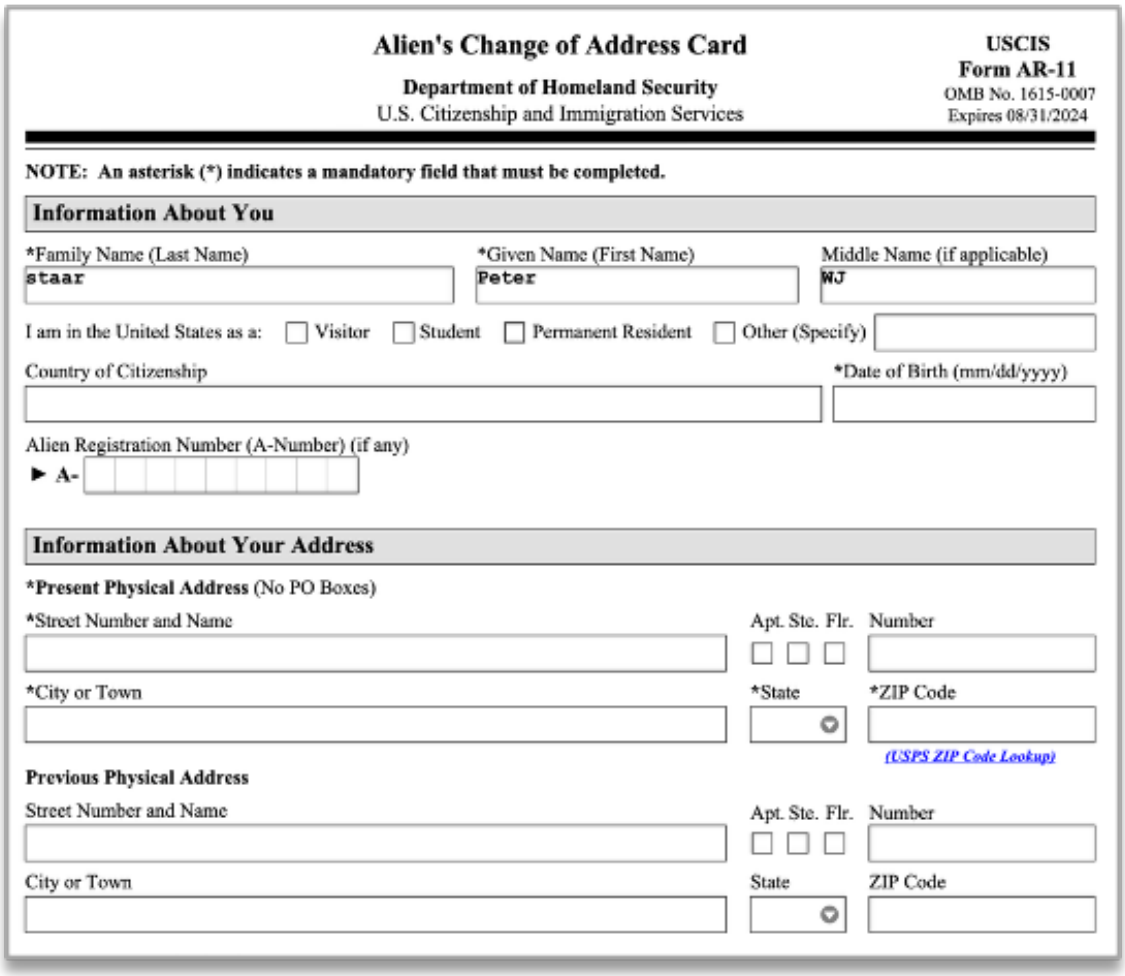
Delivery Costs:	**VAT:**
Maintenance:	**TOTAL:**

Date and time of delivery: **Dispatched by:** **Final date of delivery or implementation:** **Payment Conditions:** 60 Days from reception of correct invoice **Guarantee:**	**Delivery Supplies must be delivered together with a copy of the present purchase order. Invoices must be issued to the OHIM and send separately to the attencion of the Finance Department.**

Date of Issue and Signature of Authorising officer:	Date, Signature and Stamp for Supplier:	Date, Signature and Stamp for Acceptance:

</td></tr></table>

Use of form headings



Alien's Change of Address Card

Department of Homeland Security
U.S. Citizenship and Immigration Services

USCIS
Form AR-11
OMB No. 1615-0007
Expires 08/31/2024

NOTE: An asterisk (*) indicates a mandatory field that must be completed.

Information About You

*Family Name (Last Name): staar
*Given Name (First Name): Peter
Middle Name (if applicable): WJ

I am in the United States as a: ☐ Visitor ☐ Student ☐ Permanent Resident ☐ Other (Specify)

Country of Citizenship
*Date of Birth (mm/dd/yyyy)

Alien Registration Number (A-Number) (if any)
► A-

Information About Your Address

***Present Physical Address** (No PO Boxes)

*Street Number and Name
Apt. Ste. Flr. ☐ ☐ ☐
Number
*City or Town
*State
*ZIP Code
(USPS ZIP Code Lookup)

Previous Physical Address

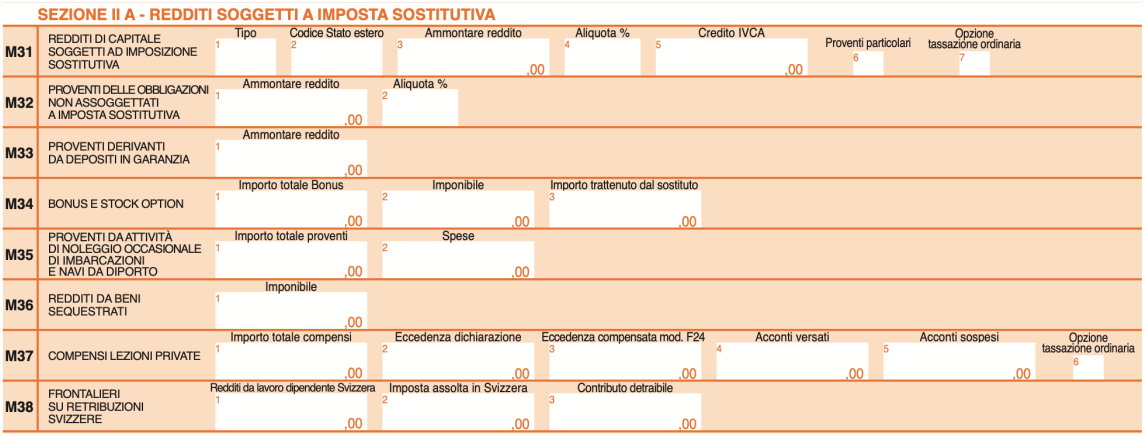
Street Number and Name
Apt. Ste. Flr. ☐ ☐ ☐
Number
City or Town
State
ZIP Code

```
<field_region>
  <field_heading>Information about you</field_heading>
  <field_item>
      <key>
          \*Family Name (Last Name)
      </key>
      <value>staar</value>
  </field_item>
  <field_item>
      <key>\*Given Name (First Name)</key>
      <value>peter</value>
  </field_item>
  <field_item>
      <key>\*Middle Name (if applicable)</key>
      <value>WJ</value>
  </field_item>
  <field_item>
      <key>I am in the United States as a:</key>
      <text><checkbox class="unselected"/> Visitor</text>
      <text><checkbox class="unselected"/> Student</text>
```

```
        <text><checkbox class="unselected"/> Permanent Resident</text>
        <text><checkbox class="unselected"/> Other (Specify)</text>
        <value></value>
    </field_item>
    <field_item>
        <key>Country of Citizenship</key>
        <value></value>
    </field_item>
    <field_item>
        <key>\*Date of Birth</key>
        <value></value>
    </field_item>
    <field_item>
        <key>Alien Registration Number (A-Number) (if any)</key>
        <value>A-</value>
    </field_item>
    <field_heading>Information About Your Address</field_heading>
    <text>\*Present Physical Address ()No Po Boxes</text>
    <field_item>
        <key>\*Street ... Name</key>
        <value></value>
    </field_item>
    <field_item>
        <key>Apt.</key>
        <checkbox class="unselected"/>
    </field_item>
    <field_item>
        <key>Ste.</key>
        <checkbox class="unselected"/>
    </field_item>
    <!--...-->

  </field_region>
```

High density form



SEZIONE II A - REDDITI SOGGETTI A IMPOSTA SOSTITUTIVA

M31	REDDITI DI CAPITALE SOGGETTI AD IMPOSIZIONE SOSTITUTIVA	1 Tipo	2 Codice Stato estero	3 Ammontare reddito ,00	4 Aliquota %	5 Credito IVCA ,00	6 Proventi particolari	7 Opzione tassazione ordinaria	
M32	PROVENTI DELLE OBBLIGAZIONI NON ASSOGGETTATI A IMPOSTA SOSTITUTIVA	1 Ammontare reddito ,00	2 Aliquota %						
M33	PROVENTI DERIVANTI DA DEPOSITI IN GARANZIA	1 Ammontare reddito ,00							
M34	BONUS E STOCK OPTION	1 Importo totale Bonus ,00	2 Imponibile ,00	3 Importo trattenuto dal sostituto ,00					
M35	PROVENTI DA ATTIVITÀ DI NOLEGGIO OCCASIONALE DI IMBARCAZIONI E NAVI DA DIPORTO	1 Importo totale proventi ,00	2 Spese ,00						
M36	REDDITI DA BENI SEQUESTRATI	1 Imponibile ,00							
M37	COMPENSI LEZIONI PRIVATE	1 Importo totale compensi ,00	2 Eccedenza dichiarazione ,00	3 Eccedenza compensata mod. F24 ,00	4 Acconti versati ,00	5 Acconti sospesi ,00	6 Opzione tassazione ordinaria		
M38	FRONTALIERI SU RETRIBUZIONI SVIZZERE	1 Redditi da lavoro dipendente Svizzera ,00	2 Imposta assolta in Svizzera ,00	3 Contributo detraibile ,00					

```
  <field_region>
      <field_heading>M31</field_heading>
      <field_heading level="2">REDDITI DI CAPITALE SOGGETTI AD IMPOSIZIONE
SOSTITUTIVA</field_heading>
```

```
      <field_item>
        <marker>1</marker>
          <key>Tipo</key>
          <value></value>
      </field_item>
      <field_item>
          <marker>2</marker>
          <key>Codice Stato estero</key>
          <value></value>
      </field_item>
      <field_item>
          <marker>3</marker>
          <key>Ammontare reddito</key>
          <value>,00</value>
      </field_item>
      <field_item>
          <marker>4</marker>
          <key>Aliquota %</key>
          <value></value>
      </field_item>
      <field_item>
          <marker>5</marker>
          <key>Credito IVCA</key>
          <value>,00</value>
      </field_item>
      <field_item>
          <marker>6</marker>
          <key>Proventi particolari</key>
          <value></value>
      </field_item>
      <field_item>
          <marker>7</marker>
          <key>Opzione tassazione ordinaria</key>
          <value></value>
      </field_item>
      <field_heading>M32</field_heading>
      <field_heading level="2">PROVENTI DELLE OBBLIGAZIONI NON ASSOGGETTATI A IMPOSTA
SOSTITUTIVA</field_heading>
      <field_item>
          <marker>1</marker>
          <key>Ammontare reddito</key>
          <value>,00</value>
      </field_item>
      <field_item>
          <marker>2</marker>
          <key>Aliquota %</key>
          <value></value>
      </field_item>
      <field_heading>M33</field_heading>
      <field_heading level="2">PROVENTI DERIVANTI DA DEPOSITI IN
GARANZIA</field_heading>
      <field_item>
          <marker>1</marker>
```

```
        <key>Ammontare reddito</key>
        <value>,00</value>
    </field_item>
    <!--...-->
</field_region>
```

Classical duality between tables and explicit key-values

10.0	Adjusted CVSS v3.1 Score			
	Vector	AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:L/A:H		
	Likelihood	Very High	Impact	Catastrophic

Affected Systems			
IP Address	**Port**	**Service**	**Version**
10.0.0.101	80/tcp, 8088/tcp	Werkzeug	3.0.1
10.0.0.102	80/tcp, 8088/tcp	Werkzeug	3.0.1
10.0.0.103	80/tcp, 8088/tcp	Werkzeug	3.0.1

```
<field_region>
    <field_item>
        <key>Adjusted CVSS v3.1 Score</key>
        <value>10.0</value>
    </field_item>
    <field_item>
        <key>Vector</key>
        <value>AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:L/A:H</value>
    </field_item>
    <field_item>
        <key>Likelihood</key>
        <value>Very High</value>
    </field_item>
    <field_item>
        <key>Impact</key>
        <value>Catastrophic</value>
    </field_item>
    <field_heading>
      Affected Systems
    </field_heading>
    <table>
    <ched/><text>IP
Address</text><ched/><text>Port</text><ched/><text>Service</text><ched/><text>Version</
text><nl/>
    <fcel/><text>10.0.0.101</text><fcel/><text>80/tcp,
8088/tcp</text><fcel/><text>Werkzeug</text><fcel/><text>3.0.1</text><nl/>
    <fcel/><text>10.0.0.102</text><fcel/><text>80/tcp,
8088/tcp</text><fcel/><text>Werkzeug</text><fcel/><text>3.0.1</text><nl/>
    <fcel/><text>10.0.0.103</text><fcel/><text>80/tcp,
8088/tcp</text><fcel/><text>Werkzeug</text><fcel/><text>3.0.1</text><nl/>
```

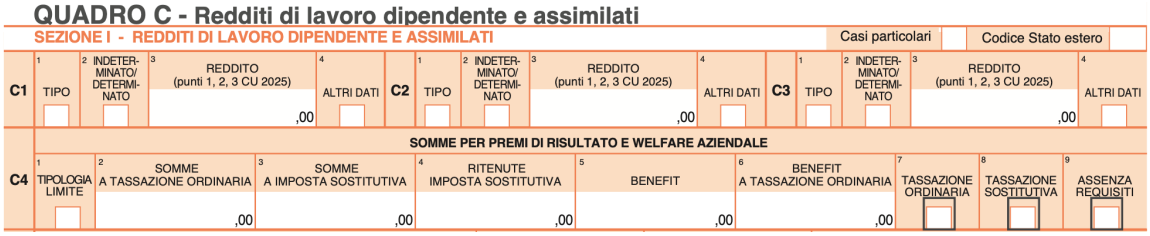
```
    </table>
</field_region>
```

Values without Keys




```
<heading>QUADRO W - Investimenti e...</heading>
<heading level="2">SEZIONE I - DATI RELATIVI...</heading>
<field_region>
    <field_heading>W1</field_heading>
    <field_item>
        <marker>1</marker>
        <key>CODICE TITOLO POSSESSO</key>
        <value></value>
    </field_item>
    <field_item>
        <marker>2</marker>
        <key>TIPO CONTRIBUENTE - IVAFE</key>
        <value></value>
    </field_item>
    <!--...-->
    <field_heading>W2</field_heading>
    <field_item>
        <marker>1</marker>
        <value></value>
    </field_item>
    <field_item>
        <marker>2</marker>
        <value></value>
    </field_item>
    <field_item>
        <marker>3</marker>
        <value></value>
    </field_item>
    <!--...-->
</field_region>
```

Another complex form deconstructed into field items



QUADRO C - Redditi di lavoro dipendente e assimilati

SEZIONE I - REDDITI DI LAVORO DIPENDENTE E ASSIMILATI | Casi particolari | Codice Stato estero

	1	2	3	4		1	2	3	4		1	2	3	4
C1	TIPO	INDETER-MINATO/ DETERMI-NATO	REDDITO (punti 1, 2, 3 CU 2025) ,00	ALTRI DATI	**C2**	TIPO	INDETER-MINATO/ DETERMI-NATO	REDDITO (punti 1, 2, 3 CU 2025) ,00	ALTRI DATI	**C3**	TIPO	INDETER-MINATO/ DETERMI-NATO	REDDITO (punti 1, 2, 3 CU 2025) ,00	ALTRI DATI

SOMME PER PREMI DI RISULTATO E WELFARE AZIENDALE

	1	2	3	4	5	6	7	8	9
C4	TIPOLOGIA LIMITE	SOMME A TASSAZIONE ORDINARIA ,00	SOMME A IMPOSTA SOSTITUTIVA ,00	RITENUTE IMPOSTA SOSTITUTIVA ,00	BENEFIT ,00	BENEFIT A TASSAZIONE ORDINARIA ,00	TASSAZIONE ORDINARIA	TASSAZIONE SOSTITUTIVA	ASSENZA REQUISITI

<table><tr><td>

```
<heading>QUADRO C - Redditi di lavoro...</heading>
<field_region>
    <field_heading>SEZIONE I - RE...</field_heading>
    <field_item>
        <key>Casi particolari</key>
        <checkbox class="unselected"/>
    </field_item>
    <field_item>
      <key>Codice Stato estero</key>
      <value></value>
    </field_item>
    <field_heading level="2">C1</field_heading>
    <field_item>
        <marker>1</marker>
        <key>TIPO</key>
        <value></value>
    </field_item>
    <field_item>
        <marker>2</marker>
        <key>INDETERMINATO/DETERMINATO</key>
        <checkbox class="unselected"/>
    </field_item>
    <field_item>
        <marker>3</marker>
        <key>REDDITO (punti 1,2,3 CU 2025)</key>
        <value>,00</value>
    </field_item>
    <field_item>
        <marker>4</marker>
        <key>ALTRI DATI</key>
        <checkbox class="unselected"/>
    </field_item>
    <field_heading level="2">C2</field_heading>
    <field_item>
        <marker>1</marker>
        <key>TIPO</key>
        <value></value>
    </field_item>
    <field_item>
        <marker>2</marker>
        <key>INDETERMINATO/DETERMINATO</key>
        <checkbox class="unselected"/>
```

```
    </field_item>
    <field_item>
        <marker>3</marker>
        <key>REDDITO (punti 1,2,3 CU 2025)</key>
        <value>,00</value>
    </field_item>
    <!--...-->
```

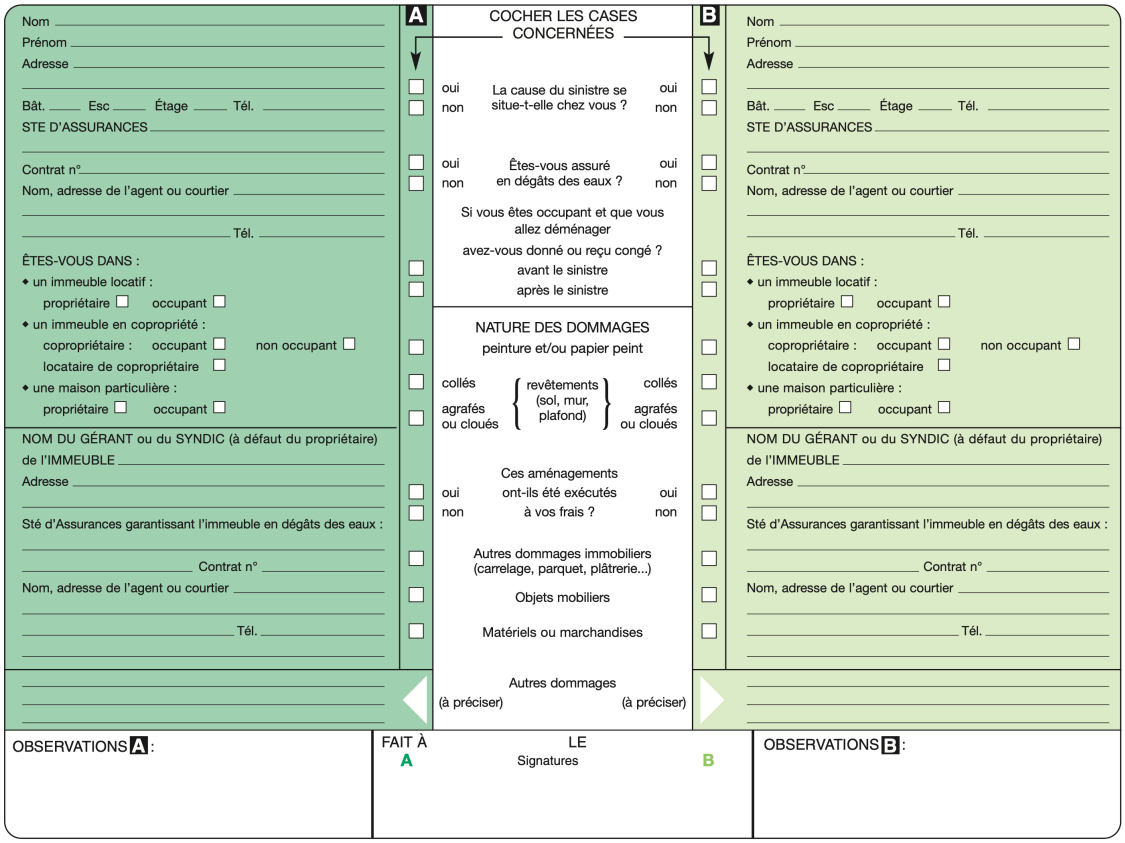
</td><td style="vertical-align: top;">

```
    <!--...-->
    <field_item>
        <marker>4</marker>
        <key>ALTRI DATI</key>
        <checkbox class="unselected"/>
    </field_item>
    <field_heading level="2">C3</field_heading>
    <field_item>
        <marker>1</marker>
        <key>TIPO</key>
        <value></value>
    </field_item>
    <field_item>
        <marker>2</marker>
        <key>INDETERMINATO/DETERMINATO</key>
        <checkbox class="unselected"/>
    </field_item>
    <field_item>
        <marker>3</marker>
        <key>REDDITO (punti 1,2,3 CU 2025)</key>
        <value>,00</value>
    </field_item>
    <field_item>
        <marker>4</marker>
        <key>ALTRI DATI</key>
        <checkbox class="unselected"/>
    </field_item>
    <field_heading level="2">C4</field_heading>
    <field_heading level="3">SOMME PER PREMI...
    </field_heading>
    <field_item>
        <marker>1</marker>
        <key>TIPOLOGIA LIMITE</key>
        <checkbox class="unselected"/>
    </field_item>
    <field_item>
        <marker>2</marker>
        <key>SOMME A TASSAZIONE ORDINARIA</key>
        <value>,00</value>
    </field_item>
    <field_item>
        <marker>3</marker>
        <key>SOMME A IMPOSTA SOSTITUTIVA</key>
```

```
        <value>,00</value>
    </field_item>
    <!--...-->
</field_region>
```

</td></tr></table>

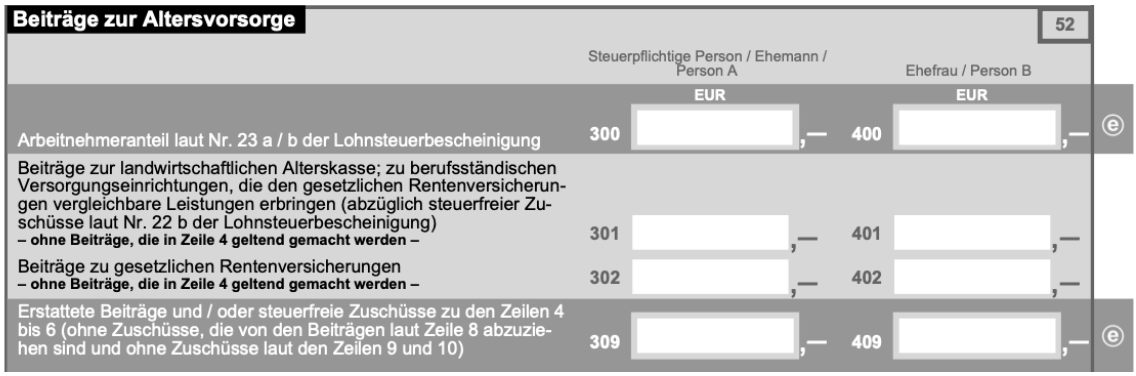
Middle section of a form with A and B choices




```
<field_region>
    <field_heading>COCHER LES CASES CONCERNEES</field_heading>
    <field_item>
        <key>La cause du sinistre se situe-t-elle chez vous ?</key>
        <text><checkbox class="unselected"/><marker>A</marker> oui</text>
        <text><checkbox class="unselected"/><marker>A</marker> non</text>
        <text><checkbox class="unselected"/><marker>B</marker> oui</text>
        <text><checkbox class="unselected"/><marker>B</marker> non</text>
    </field_item>
    <field_item>
        <key>Êtes-vous assuré en dégâts des eaux ?</key>
        <text><checkbox class="unselected"/><marker>A</marker> oui</text>
        <text><checkbox class="unselected"/><marker>A</marker> non</text>
        <text><checkbox class="unselected"/><marker>B</marker> oui</text>
        <text><checkbox class="unselected"/><marker>B</marker> non</text>
    </field_item>
    <field_item>
```

```
          <key>Si vous êtes occupant et que vous allez déménager avez-vous donné ou
reçu congé ?</key>
          <text><checkbox class="unselected"/><marker>A</marker> avant le
sinistre</text>
          <text><checkbox class="unselected"/><marker>A</marker> après le
sinistre</text>
          <text><checkbox class="unselected"/><marker>B</marker> avant le
sinistre</text>
          <text><checkbox class="unselected"/><marker>B</marker> après le
sinistre</text>
      </field_item>
      <field_heading>NATURE DES DOMMAGES peinture et/ou papier peint</field_heading>
      <field_item>
          <key>revêtements (sol, mur, plafond)</key>
          <text><checkbox class="unselected"/><marker>A</marker> collés</text>
          <text><checkbox class="unselected"/><marker>A</marker> agrafés ou
cloués</text>
          <text><checkbox class="unselected"/><marker>B</marker> collés</text>
          <text><checkbox class="unselected"/><marker>B</marker> agrafés ou
cloués</text>
      </field_item>
      <field_item>
          <key>Ces aménagements ont-ils été exécutés à vos frais ?</key>
          <text><checkbox class="unselected"/><marker>A</marker> oui</text>
          <text><checkbox class="unselected"/><marker>A</marker> non</text>
          <text><checkbox class="unselected"/><marker>B</marker> oui</text>
          <text><checkbox class="unselected"/><marker>B</marker> non</text>
      </field_item>
      <field_item>
          <key>Autres dommages immobiliers (carrelage, parquet, plâtrerie...)</key>
          <text><checkbox class="unselected"/><marker>A</marker></text>
          <text><checkbox class="unselected"/><marker>B</marker></text>
      </field_item>
      <field_item>
          <key>Objets mobiliers</key>
          <text><checkbox class="unselected"/><marker>A</marker></text>
          <text><checkbox class="unselected"/><marker>B</marker></text>
      </field_item>
      <field_item>
          <key>Matériels ou marchandises</key>
          <text><checkbox class="unselected"/><marker>A</marker></text>
          <text><checkbox class="unselected"/><marker>B</marker></text>
      </field_item>
      <field_item>
          <key>Autres dommages</key>
          <value><marker>A</marker><hint>(à préciser)</hint></value>
          <value><marker>B</marker><hint>(à préciser)</hint></value>
      </field_item>
  </field_region>
```

Tabular form with strong 2D value relationship



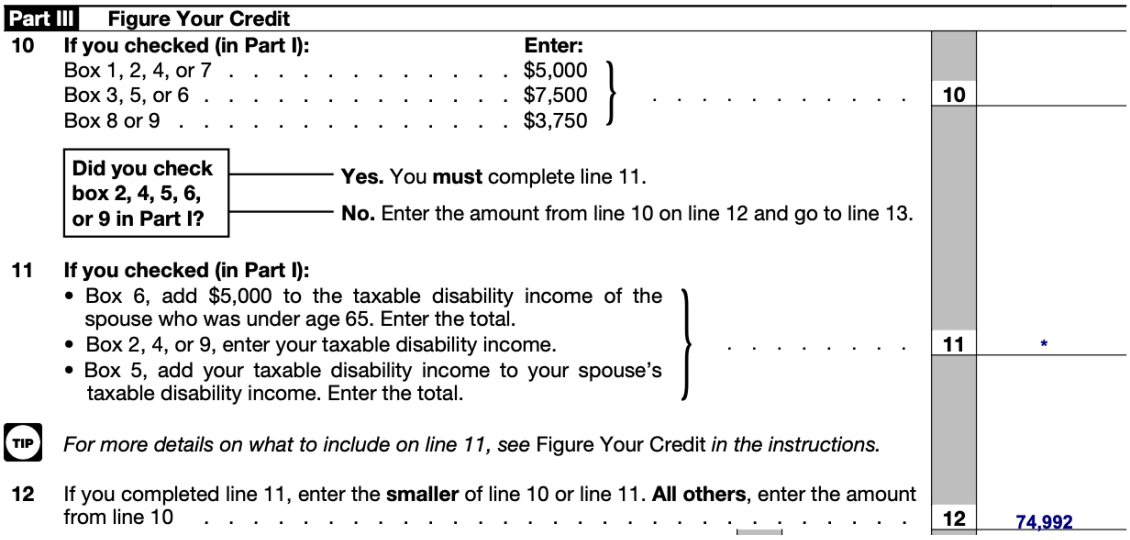
Beiträge zur Altersvorsorge 52

		Steuerpflichtige Person / Ehemann / Person A		Ehefrau / Person B		
			EUR		EUR	
4	Arbeitnehmeranteil laut Nr. 23 a / b der Lohnsteuerbescheinigung	300	,—	400	,—	ⓔ
5	Beiträge zur landwirtschaftlichen Alterskasse; zu berufsständischen Versorgungseinrichtungen, die den gesetzlichen Rentenversicherungen vergleichbare Leistungen erbringen (abzüglich steuerfreier Zuschüsse laut Nr. 22 b der Lohnsteuerbescheinigung) **– ohne Beiträge, die in Zeile 4 geltend gemacht werden –**	301	,—	401	,—	
6	Beiträge zu gesetzlichen Rentenversicherungen **– ohne Beiträge, die in Zeile 4 geltend gemacht werden –**	302	,—	402	,—	
7	Erstattete Beiträge und / oder steuerfreie Zuschüsse zu den Zeilen 4 bis 6 (ohne Zuschüsse, die von den Beiträgen laut Zeile 8 abzuziehen sind und ohne Zuschüsse laut den Zeilen 9 und 10)	309	,—	409	,—	ⓔ

```
  <field_region>
    <table>
    <srow/><text>Beiträge zur Altersvorsorge</text>
<srow/>
<srow/><text>52</text>                  <ecel/><nl/>
    <ecel/>
<ched/><text>Steuerpflichtige Person / Ehemann / Person A</text>  <ched/><text>Ehefrau
/ Person B</text>  <ecel/><nl/>
    <fcel/><text>Arbeitnehmeranteil laut Nr. 23 a / b der
Lohnsteuerbescheinigung</text>  <fcel/><text>*FORM1*,-</text>
<fcel/><text>*FORM2*,-</text>           <fcel/><text>e</text><nl/>
    <fcel/><text>Beiträge zur landwirtschaftlichen Alterskasse; zu
berufsständ...</text>  <fcel/><text>*FORM3*,-</text>
<fcel/><text>*FORM4*,-</text>           <ecel/><nl/>
    <fcel/><text>Beiträge zu gesetzlichen Rentenversicherungen...</text>
<fcel/><text>*FORM5*,-</text>
<fcel/><text>*FORM6*,-</text>           <ecel/><nl/>
    <fcel/><text>Erstattete Beiträge und / oder steuerfreie Zuschüsse zu den...</text>
<fcel/><text>*FORM7*,-</text>
<fcel/><text>*FORM8*,-</text>           <fcel/><text>e</text><nl/>
    <!--...-->
    </table>
  </field_region>
  <!--...-->
```

FORMS referred above: FORM1*:*
<field_item><key>300</key><value></value><hint>EUR</hint></field_item> FORM2*:*
<field_item><key>400</key><value></value><hint>EUR</hint></field_item> FORM4*:*
<field_item><key>401</key><value></value></field_item> FORM5*:*
<field_item><key>302</key><value></value></field_item> FORM3*:*
<field_item><key>301</key><value></value></field_item> FORM6*:*
<field_item><key>402</key><value></value></field_item> FORM7*:*
<field_item><key>309</key><value></value></field_item> FORM8*:
<field_item><key>409</key><value></value></field_item>

Mix table and form elements



Part III **Figure Your Credit**

10 **If you checked (in Part I):** **Enter:**

Box 1, 2, 4, or 7	$5,000
Box 3, 5, or 6	$7,500
Box 8 or 9	$3,750

10

Did you check box 2, 4, 5, 6, or 9 in Part I?

Yes. You **must** complete line 11.

No. Enter the amount from line 10 on line 12 and go to line 13.

11 **If you checked (in Part I):**

- Box 6, add $5,000 to the taxable disability income of the spouse who was under age 65. Enter the total.
- Box 2, 4, or 9, enter your taxable disability income.
- Box 5, add your taxable disability income to your spouse's taxable disability income. Enter the total.

11 *

TIP *For more details on what to include on line 11, see* Figure Your Credit *in the instructions.*

12 If you completed line 11, enter the **smaller** of line 10 or line 11. **All others**, enter the amount from line 10 . . . . . 12 74,992

```
<!--...-->
<heading>Part III</heading>
<text>Figure Your Credit</text>
<text>10</text>
<table>
  <ched/><text>If you checked (in Part l):</text><ched/><text>Enter</text><nl/>
  <fcel/><text>Box 1, 2, 4, or 7</text><fcel/><text>$5,000</text><nl/>
  <fcel/><text>Box 3, 5, or 6</text><fcel/><text>$7,500</text><nl/>
  <fcel/><text>Box 8 or 9</text><fcel/><text>$3,750</text><nl/>
</table>
<field_region><field_item><key>10</key><value></value></field_item></field_region>
<text>11 If you checked (in Part I):</text>
<list>
    <ldiv/>
    <text>Box 6, add $5,000 to the taxable...</text>
    <ldiv/>
    <text>Box 2, 4, or 9, enter your taxable...</text>
    <ldiv/>
    <text>BBox 5, add your taxable disabilit...</text>
</list>
<field_region><field_item><key>11</key><value>.</value></field_item></field_region>
<picture><label value="logo"/></picture>
<text>For more details on what to include on line 11...</text>
<text>12 If you completed line 11, enter the smaller...</text>

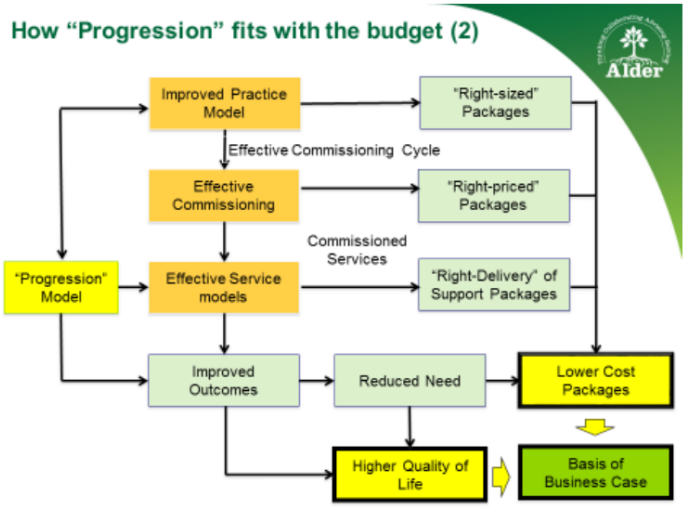
<field_region><field_item><key>12</key><value>74,992</value></field_item></field_region
>
<!--...-->
```

16. Learning Disabilities and Autism

Population Profile

We support in the region of 680 people with a learning disability through social care. 192 of those live in residential care, a proportion higher than in other areas**.** The number of people with a learning disability is expected to rise by 30 people each year as life expectancy rises and a larger number of young people become the responsibility of adult social care each year.

Swindon has a higher number of people with a learning disability living in residential care and high costs supported housing. Our analysis of social work assessments and plans has also found that there is a need to be outcome and progression focused so that people learn new skills. A pilot has been undertaken with Wiltshire, with consultant Alder, on the implementation of progression planning.



Key to this work has been the higher quality of life, through delivery of sustainable outcomes, and also lower cost packages. With both of these comes the opportunity to innovate across the market.

Population Projection	2015	2020	2025	2030
People aged 18-64 predicted to have a learning disability	3,344	3,451	3,529	3,585
People aged 65+ predicted to have a learning disability	687	797	923	1087

Source: www.pansi.org.uk and www.poppi.org.uk (based on national not local population projections).

```
<!--...-->
<field_region>
  <field_item>
    <key>Source</key>
    <value>www.pansi.org.uk and ... projections).</value>
  </field_item>
</field_region>
<!--...-->
```

Detailed examples can be seen here: Form Examples

Tracks

A `<track>` captures a time-aligned media transcript — subtitles, captions, or diarized speech associated with an audio or video recording. It is a semantic element and may begin with an element head.

The body of a `<track>` may begin with a single `<cover>` — a representative image for the track as a whole (podcast artwork, a poster, a title card). It has the same shape as `<frame>` (an optional element head followed by an optional `<src>`).

The rest of the body is a sequence of *cue blocks*. Each cue block is introduced by a `<bdiv/>` delimiter (analogous to `<ldiv/>` for lists): the first cue block must be a `<bdiv/>`, and a cue block spans everything between two sibling `<bdiv/>` elements (or until `</track>`).

A cue block consists of, in order:

- a **start time** (mandatory): a run of `<hours value="H"/>`, `<minutes value="M"/>`, `<seconds value="S"/>`, `<msecs value="MS"/>` in that order. Only `<seconds>` is required; `<hours>`, `<minutes>` and `<msecs>` each default to `0` when omitted. `<seconds>` anchors the run, so two consecutive runs stay unambiguous.
- an **end time** (optional): the same run shape, immediately following the start time. A cue block covers the inclusive interval `[start, end]`. When the end time is omitted it is taken to equal the start time, i.e. the cue block is the single instant `[start, start]` — for example, the timestamp of a `<frame>` or a point annotation.
- an optional `<chapter>` — a chapter or section title (see Chapters below), handled like `<text>`: it may carry its own element head and inline formatting.
- an optional `<frame>` — a still image for the cue block's start time, with an optional element head and an optional `<src>`. Typically a video frame, but equally a slide, a keyframe, or any representative still for that moment. A point sample; valid on any cue block. (For a track-wide image, use `<cover>` instead.)
- an optional `<audio>` — the recording over the cue block's interval `[start, end]`, with the same shape as `<frame>`. It is a temporal crop of the track's audio, the way a `<picture>` crop corresponds to its page region; the timestamps are authoritative and the clip is a best-effort fragment. Requires the cue block to carry an end time.
- an optional **transcript**: a sequence of speaker turns. A turn is an optional `<voice>` (the speaker attribution, which may itself be styled) followed by the spoken text with inline

formatting; it runs until the next `<voice>` or the end of the cue block. An empty `<voice/>` ends the current speaker's turn — the text after it is unattributed. How unattributed text is interpreted — e.g. as an unknown speaker, no speaker (e.g. a sound description), continuation of the last named speaker, etc. — is application-defined.

Minimal cue block (start time and one line of text) — `<hours>`, `<minutes>` and `<msecs>` omitted:

```
<track>
  <bdiv/>
  <seconds value="2"/>
  Hello, and welcome.
</track>
```

Start and end time, two attributed turns:

```
<track>
  <bdiv/>
  <minutes value="0"/><seconds value="2"/><msecs value="500"/>
  <minutes value="0"/><seconds value="7"/>
  <voice>John</voice>Hi Mary!
  <voice><italic>Mary</italic></voice><bold>Hello</bold> John!
</track>
```

Cue block with a video frame and an audio clip:

```
<track>
  <bdiv/>
  <minutes value="1"/><seconds value="4"/>
  <minutes value="1"/><seconds value="9"/>
  <frame><src uri="assets/frames/00-01-04.jpg"/></frame>
  <audio><src uri="assets/clips/00-01-04.opus"/></audio>
  <voice>Narrator</voice>The results are shown on screen.
</track>
```

Per-cue `<frame>` and `<audio>` reference standalone fragments (a still, a clip) — typically archive `assets/` or a `data:` URI. The complete recording a track's timestamps refer to is carried separately, as `audio/{N}.*` / `video/{N}.*` in a DocLang archive.

Crossing the hour mark, and an unattributed turn:

```
<track>
  <label value="lecture"/>
  <bdiv/>
  <minutes value="59"/><seconds value="58"/><msecs value="450"/>
  <hours value="1"/><minutes value="0"/><seconds value="1"/><msecs value="210"/>
  So that concludes the first part.
  Take a five minute break.
</track>
```

The kind of a track —for example whether it is audio-only or video— is not modelled explicitly. A producer that wants to record it may do so informally with a `<label>` in the

track's element head (`audio`, `video`, `podcast`, `screencast` etc.); label values are open and not validated. For instance:

```
<track>
  <label value="audio"/>
  <bdiv/>
  <seconds value="0"/>
  <seconds value="2"/>
  <voice>Host</voice>
  Welcome back to the show.
</track>
```

Chapters

A cue block may carry a `<chapter>` — a chapter or section title — placed just after its timestamps (before any `<frame>`). A `<chapter>` **marks a boundary at its cue block's start time and nothing else**: the chapter runs from that instant until the next cue block that carries a `<chapter>` (or `</track>`). The cue block's own end time bounds its transcript and audio, not the chapter. The stretch before the first `<chapter>` is unchaptered.

Chapters therefore form a **flat partition** of the timeline: no nesting, no overlap, no explicit chapter end. "Which chapter is active at time *T*?" is answered by the last `<chapter>` at or before *T*.

```
<track>
  <bdiv/>
  <minutes value="0"/><seconds value="0"/>
  <chapter>Introduction</chapter>
  Welcome, everyone.
  <bdiv/>
  <minutes value="0"/><seconds value="40"/>
  Still in the introduction; this cue block starts no chapter.
  <bdiv/>
  <minutes value="1"/><seconds value="30"/>
  <chapter><italic>Part 1</italic> — Background</chapter>
  Let's start with some history.
</track>
```

Because only the start time counts, a chapter boundary may fall inside an earlier, still-open cue block. Given a cue block with `<chapter>A</chapter>` starting at `01:00` and one with `<chapter>B</chapter>` starting at `01:15`, chapter A is `[01:00, 01:15)` and B begins at `01:15` — even if A's cue block carries transcript out to `01:30`.

Chapters MUST begin at strictly increasing times — two chapters cannot mark the same instant. A `<chapter>` has no attributes in this version; nesting would be a future `level` attribute.

Notes:

- A `<track>` body may open with one `<cover>` (a track-wide image); the first cue block, and every non-empty `<track>` body, then begins with a `<bdiv/>`. There must be no text before the first cue block, nor between a `<bdiv/>` and its start time.
- Every timestamp run carries `<seconds>` (the only required component); an end time, when present, must not be earlier than the start time. Cue-block intervals are inclusive of both endpoints; a missing end time means the interval `[start, start]`.
- Cue blocks appear in non-decreasing order of start time. They may still overlap — for example, two speakers talking at once. When two cue blocks share a start time, their relative order is not further constrained (end times are not used as a tie-breaker).
- A `<frame>` is the still at the cue block's start time. An `<audio>` clip is the recording over `[start, end]`, so a cue block with an `<audio>` must have an end time. Its duration is expected to correspond to `end - start`, but small differences from codec framing and encoder padding are normal and not significant; the timestamps, not the clip, are authoritative, and DocLang does not decode media to check this.
- A `<chapter>`, when present, comes after the timestamps and before any `<frame>`. It defines a chapter boundary at the cue block's start time only — see Chapters.
- `<msecs value="…"/>` accepts any integer in `[0, 999]`.
- `<cover>`, `<bdiv>`, `<chapter>`, `<frame>`, `<audio>`, `<voice>`, and the timestamp elements are only meaningful inside a `<track>`.

Split structure

We can capture content that is split (e.g. across columns or across pages) using the `<thread thread_id="N"/>` element, where `N` is a unique identifier.

The basic structure is shown below, e.g. for a `text` tag:

```
<text>
  <thread thread_id="1"/>
  This text item starts here
</text>
<!--...-->
<text>
  <thread thread_id="1"/>
  and continues here.
</text>
```

Cross-column structure example

Each block that has location information is a top-level tag of the corresponding label, e.g. "text".

Elements which belong to the same document component must have the same thread ID and same host element type (e.g. all under `<text>`, not mixed `<text>` and `<picture>`).

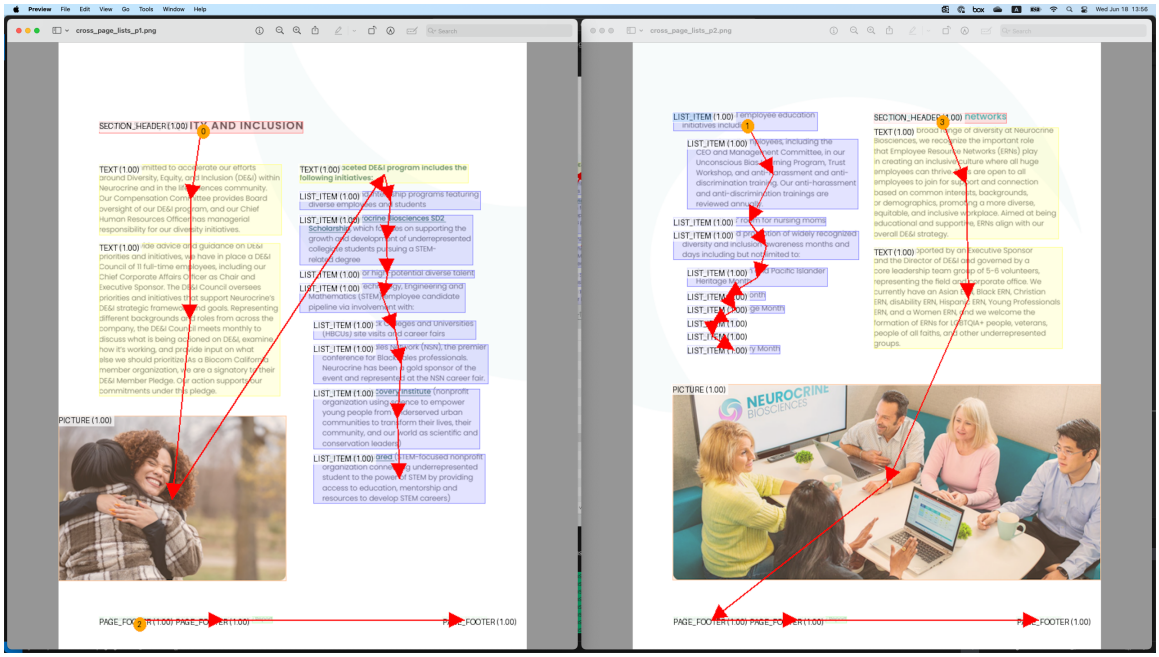
```
<!-- ... -->
<text>
```

```
  <thread thread_id="1"/>
  <location value="10"/><location value="20"/>
  <location value="30"/><location value="40"/>
  where τ<subscript>x,y,z</subscript> are the Pauli matrices acting
  on Nambu space. We consider a circular-shaped boundary, the nor-
</text>

<picture>
  <caption>
    <location value="20"/><location value="30"/>
    <location value="40"/><location value="50"/>
    FIG. 3. The modules of the inner product of two MES spinors
    <formula><!-- ... --></formula>
    <!-- ... -->
  </caption>
  <!-- ... -->
</picture>

<text>
  <thread thread_id="1"/>
  <location value="30"/><location value="40"/>
  <location value="50"/><location value="60"/>
  mal direction of the boundary tangent for arbitrary angle θ is
  <formula><!-- ... --></formula>
  <!-- ... -->
</text>
```

Cross-page structure example



The scenario in the above figure is represented as follows:

```
<!--...-->
<text>
```

```
    <location value="10"/><location value="20"/>
    <location value="30"/><location value="40"/>
    <![CDATA[Our multi-faceted DE&I program includes the following initiatives:]]>
</text>

<list>
    <thread thread_id="1"/>
    <ldiv/>
    <text>
        <location value="15"/><location value="25"/>
        <location value="35"/><location value="45"/>
        Mentorships and internship programs featuring diverse employees and students
    </text>
    <!--...-->
    <ldiv/>
    <text>
        <location value="20"/><location value="30"/>
        <location value="40"/><location value="50"/>
        Build Science, Technology, Engineering and Mathematics (STEM) employee
candidate pipeline via involvement with:
        <list>
            <ldiv/>
            <text>
                <location value="25"/><location value="35"/>
                <location value="45"/><location value="55"/>
                Historically Black Colleges and Universities (HBCUs) site visits and
career fairs
            </text>
            <!--...-->
            <ldiv/>
            <text>
                <location value="30"/><location value="40"/>
                <location value="50"/><location value="60"/>
                San Diego Squared (STEM-focused nonprofit organization connecting
underrepresented student to the power
                of STEM by providing access to education, mentorship and resources to
develop STEM careers)
            </text>
        </list>
    </text>
</list>

<page_footer>
    <location value="35"/><location value="45"/>
    <location value="55"/><location value="65"/>
    16 Neurocrine Biosciences
</page_footer>

<page_break/>

<list>
    <thread thread_id="1"/>
    <ldiv/>
```

```
    <text>
        <location value="40"/><location value="50"/>
        <location value="60"/><location value="70"/>
        <![CDATA[Build upon DE&I employee education initiatives including: ...]]>
    </text>
    <!--...-->
</list>
<!--...-->
```

References

Cross-references

Cross-references can be captured using the `<xref>` element as part of the element head, for pointing to a thread based on the `thread_id`.

Inline cross-reference example

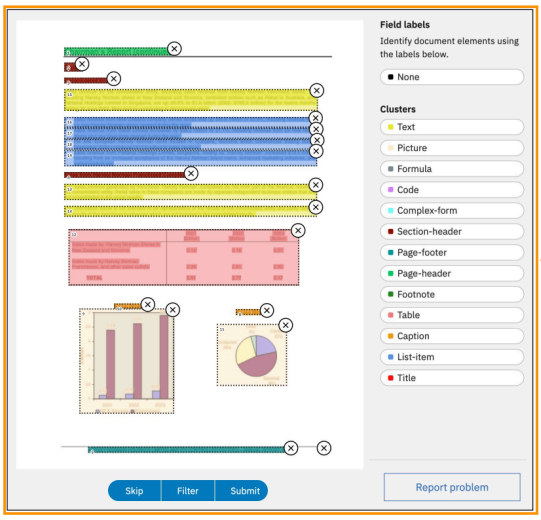


Figure 3: Corpus Conversion Service annotation user interface. The PDF page is shown in the background, with overlaid text-cells (in darker shades). The annotation boxes can be drawn by dragging a rectangle over each segment with the respective label from the palette on the right.

include publication repositories such as arXiv[3], government offices, company websites as well as data directory services for financial reports and patents. Scanned documents were excluded wherever possible because they can be rotated or skewed. This would not allow us to perform annotation with rectangular bounding-boxes and therefore complicate the annotation process.

Preparation work included uploading and parsing the sourced PDF documents in the Corpus Conversion Service (CCS) [22], a cloud-native platform which provides a visual annotation interface and allows for dataset inspection and analysis. The annotation interface of CCS is shown in Figure 3. The desired balance of pages between the different document categories was achieved by selective subsampling of pages with certain desired properties. For example, we made sure to include the title page of each document and bias the remaining page selection to those with figures or tables. The latter was achieved by leveraging pre-trained object detection models from PubLayNet, which helped us estimate how many figures and tables a given page contains.

Phase 2: Label selection and guideline. We reviewed the collected documents and identified the most common structural features they exhibit. This was achieved by identifying recurrent layout elements and lead us to the definition of 11 distinct class labels. These 11 class labels are *Caption*, *Footnote*, *Formula*, *List-item*, *Page-footer*, *Page-header*, *Picture*, *Section-header*, *Table*, *Text*, and *Title*. Critical factors that were considered for the choice of these class labels were (1) the overall occurrence of the label, (2) the specificity of the label, (3) recognisability on a single page (i.e. no need for context from previous or next page) and (4) overall coverage of the

```
<picture>
  <thread thread_id="1"/>
  <!--...-->
</picture>
<!--...-->
<text>
  ... is shown in <text><xref thread_id="1"/>Figure 3</text>. The desired balance...
</text>
```

Hyperlinks

Hyperlinks can be captured using the `<href>` element as part of the element head, for pointing to a URI.

Basic example:

```
<text>
  Visit our <text><href uri="https://www.example.com"/>website</text> for more
information.
</text>
```

Hyperink from a non-text element:

```
<text>Click on the image below:</text>
<picture>
  <href uri="https://www.example.com"/>  <!-- the target URI -->
  <src uri="https://www.example.com/image.jpg"/>  <!-- the image source URI -->
</picture>
```

Formatting

Formatting may be preserved through nested tags or escape sequences:

- Bold, italic, underline, strikethrough, handwriting
- Superscript, subscript
- Text direction markers

Examples of basic formatting:

```
<text>
  This text contains <bold>bold</bold>, <italic>italic</italic>,
  <underline>underlined</underline>, and <strikethrough>struck-through</strikethrough>
text.
</text>
```

Nested formatting:

```
<text>
  You can combine <bold><italic>bold and italic</italic></bold> or
  <underline><bold>underlined and bold</bold></underline> text.
</text>
```

Handwriting formatting (for handwritten annotations or text):

```
<text>
  The printed text says "Sign here:" followed by
  <handwriting>John Smith</handwriting> in handwritten form.
</text>
```

Superscript and subscript:

```
<text>
  1<superscript>st</superscript> question: What is the formula for water?
  Answer: H<subscript>2</subscript>O.
</text>
```

Page break with continuation

Page breaks are complex components that interrupt the flow of a document. They can interrupt paragraphs, tables, lists, etc. In general, we follow two rules,

20. If content spans across one (or more) page breaks, add `<thread thread_id="N"/>` to each fragment, reusing the same `thread_id`.
21. For the follow up content of the page, we follow a reading order and close all open elements before the `<page_break/>` element is introduced.

An easy example is below,

```
<doclang>
  <text>
    <thread thread_id="1"/>
    <caption>Some caption</caption>
    This paragraph spans across
  </text>
  <page_break/>
  <text><thread thread_id="1"/>multiple pages.</text>
</doclang>
```

Often, we have more complicated page breaks, in which a (nested) list is split across pages and further interrupted by other semantic elements (think page footers). In this case, we demand that all elements of the first page are added and/or closed **before** the page break and then opened again in the appropriate way after the page break, with the intent that the content in between the page breaks is a valid DocLang tree.

A more complicated example is shown below in which we break the content of a list-item,

```
<doclang>
  <list class="ordered">
    <thread thread_id="1"/>
    <ldiv/>
    <text>First item</text>
    <ldiv/>
    <text><thread thread_id="2"/>Second </text>
    <!--...-->
  </list>
  <page_footer>...</page_footer>
  <page_break/>
  <list class="ordered">
    <thread thread_id="1"/>
    <ldiv/>
    <text><thread thread_id="2"/>item</text>
  </list>
  <!--...-->
</doclang>
```

Above, `<thread thread_id="1"/>` captures that the list itself is split, while `<thread thread_id="2"/>` captures that a particular list item is split.

Custom vocabularies

The examples below illustrate custom vocabulary use, namely for capturing a chemistry picture in simplified molecular-input line-entry system (SMILES) representation.

For details, see custom metadata and vocabulary guidelines in Recommendations.

For shared/interoperable documents, using a formal XML namespace is recommended:

```
<picture>
  <label value="chemistry_structure"/>
  <custom xmlns:acme="https://example.com/ns/doclang/custom/chemistry/1">
    <acme:smiles>C1=CC=C(C=C1)C(=O)O</acme:smiles>
  </custom>
  <src uri="molecule.svg"/>
</picture>
```

For local/private usage where formal namespaces are not used, a collision-resistant project prefix can be used:

```
<picture>
  <label value="chemistry_structure"/>
  <custom>
    <acme_smiles>C1=CC=C(C=C1)C(=O)O</acme_smiles>
  </custom>
  <src uri="molecule.svg"/>
</picture>
```

Annexes

Annex A (normative): Reference

Special elements

This category comprises elements with specialized document-level function.

<doclang>

The document root element. Starts with an optional `<head>` followed by a sequence of applicable elements.

Allowed context

Exists exactly once, as root element.

Attributes

Attribute	Required / Optional	Allowed values	Description
`xmlns`	Optional; default: "https://www.doclang.ai/ns/v0"	{"https://www.doclang.ai/ns/v0"}	The DocLang specification version namespace.
`version`	Optional; if omitted, the version is unspecified	"MAJOR.MINOR" of a published DocLang specification version	The DocLang specification version the DocLang instance is supposed to validate against, in "MAJOR.MINOR" format, i.e. first two positions of Semantic Versioning. If omitted, the version is unspecified (see Omitted version).

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Not allowed
Primary semantic elements	Allowed

Example

```
<doclang>
  <!-- content -->
</doclang>
```

<head>

Includes doc-level metadata.

Allowed context

Can only be first child of `<doclang>`.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Not allowed
Primary semantic elements	Not allowed

<page_break>

Indicates a page break. A paginated document may be divided into pages using the `<page_break/>` empty element. Any page content, as split by `<page_break/>`, forms a valid DocLang document body, i.e. would be a valid DocLang instance if wrapped in a `doclang` root element.

Allowed context

Can only be child of `<doclang>`.

Attributes

None

Allowed content types

None (empty element).

Example

```
<doclang>
  <!-- first page content -->
  <page_break/>
  <!-- second page content -->
</doclang>
```

Semantic elements

Semantic elements capture core document components with specific meaning and functional role (e.g. a paragraph, a table, a list etc.) and may optionally begin with a element head. They are generally meant to be read as separate blocks, such as a paragraph, although some can also be nested within text. Semantic elements that can appear on the top level within

`<doclang>` are called *primary*, while those that can only appear within other semantic elements are called *secondary*.

<text>

Represents a piece of cohesive text as that would appear in a paragraph. A list item, or a cell of a `<table>`, `<index>` or `<tabular>`, may also consist of raw text, optionally preceded by an element head, without wrapping `<text>` tags. Such content is called a *virtual `<text>`* and is interpreted exactly as if it were wrapped in `<text>` tags.

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<heading>

Captures a document heading.

Allowed context

Any context that allows semantic elements.

Attributes

Attribute	Required / Optional	Allowed values	Description
`level`	Optional; default "1"	Positive integer	The heading depth (1 = top-level).

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<footnote>

Captures footnote content.

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<page_header>

Captures page header content (material repeated at the top of a page).

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<page_footer>

Captures page footer content (material repeated at the bottom of a page).

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<field_region>

Serves for scoping of field items, for example encapsulating a whole form.

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed

Element head	Allowed
Raw text	Not allowed
Primary semantic elements	Allowed

<list>

Captures a list. List items are started by the respective structural elements (`<ldiv>`). A non-empty `<list>` element body must begin with such a structural element. A list item can be any semantic element sequence or a virtual `<text>`.

Allowed context

Any context that allows semantic elements.

Attributes

Attribute	Required / Optional	Allowed values	Description
`class`	Optional; default: "unordered"	{"unordered", "ordered"}	The list type.

Allowed content types

Content type	Allowed / Not allowed
Element head	TRUE, also on cell level in case of virtual `<text>`
Raw text	Only on cell level, in case of virtual `<text>`
Primary semantic elements	Allowed

<table>

Captures a table in an OTSL-based format. Table cells are started by the respective structural elements (`<fcel>` etc). A non-empty `<table>` element body must begin with such a structural element. A table cell can be any semantic element sequence or a virtual `<text>`.

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	TRUE, also on cell level in case of virtual `<text>`
Raw text	Only on cell level, in case of virtual `<text>`
Primary semantic elements	Allowed

<index>

Captures an index, e.g. for a table of contents or glossary, in an OTSL-based format. Index cells are started by the respective structural elements (`<fcel>` etc). A non-empty `<index>`

element body must begin with such a structural element. A cell can be any semantic element sequence or a virtual `<text>`.

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	TRUE, also on cell level in case of virtual `<text>`
Raw text	Only on cell level, in case of virtual `<text>`
Primary semantic elements	Allowed

<formula>

Raw LaTeX formula content, i.e. without any LaTeX-specific wrapping such as `$ ... $`, `$$ ... $$`, `\( ... \)`, `\[ ... \]`, `\begin{math} ... \end{math}` or `\begin{equation} ... \end{equation}`.

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<code>

Captures a code snippet, either as a standalone block or inlined within a semantic element. For language classification, use a `<label>`in the element head (see Recommendations).

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed

Primary semantic elements	Not allowed

<picture>

The element body begins with a picture-specific sequence, followed by content allowed in any other semantic element body. The picture-specific sequence is:

- an optional `<src>` element
- an optional `<tabular>` element (only allowed for `<picture class="chart">`)

Allowed context

Any context that allows semantic elements.

Attributes

Attribute	Required / Optional	Allowed values	Description
`class`	Optional; default: "undefined"	{"undefined", "chart"}	The picture type.

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Not allowed
Primary semantic elements	Allowed

<marker>

Captures the visible glyph or identifier (e.g. number) of a marker, e.g. in the context of a list item.

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<group>

Container for encapsulating multiple semantic elements.

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Not allowed
Primary semantic elements	Allowed

<field_heading>

Field heading within a `<field_region>; analogous to `<heading>` but scoped to field structures.

Allowed context

Can only be descendant of `<field_region>`.

Attributes

Attribute	Required / Optional	Allowed values	Description
`level`	Optional; default "1"	Positive integer	The field heading depth (1 = top-level).

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<field_item>

Scoping of a field key (optional) and any corresponding values.

Allowed context

Can only be descendant of `<field_region>`.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Not allowed
Primary semantic elements	Allowed

<key>

The key of a field (may correspond to 0-N field values).

Allowed context

Can only be descendant of `<field_item>`.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<value>

A value of a field (may correspond to 0 or 1 field key).

Allowed context

Can only be descendant of `<field_item>`.

Attributes

Attribute	Required / Optional	Allowed values	Description
`class`	Optional; default: "read_only"	{"read_only", "fillable"}	The value type.

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<hint>

A hint regarding a field.

Allowed context

Can only be descendant of `<field_region>`.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<caption>

Optional part of the element head for capturing an associated caption. Unlike `<description>` or `<summary>`, `<caption>` is an actual document component, which can have its own location information etc. For example, a caption shown underneath a chart. An element head may contain multiple captions, e.g. a figure with both a title above and a note below, or the same caption in several languages.

Allowed context

Can only be part of the element head of a semantic element.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<track>

Captures a time-aligned media transcript (subtitles, captions, or diarized speech). The body may open with a single `<cover>` (a track-wide image), then a sequence of cue blocks, each introduced by a `<bdiv>` delimiter; a non-empty `<track>` body must begin with a `<cover>` or a `<bdiv>`. A cue block carries a mandatory start time, an optional end time (each a run of `<hours>`/`<minutes>`/`<seconds>`/`<msecs>`), an optional `<chapter>`, an optional `<frame>`, an optional `<audio>`, and an optional transcript of `<voice>`-attributed turns.

Allowed context

Any context that allows semantic elements.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Only within a cue block transcript
Primary semantic elements	`<cover>` before the first cue block; within cue blocks, `<chapter>`, `<frame>`, `<audio>`, and `<voice>`

Example

```
<doclang>
  <track>
    <bdiv/>
    <minutes value="0"/><seconds value="2"/><msecs value="500"/>
    <minutes value="0"/><seconds value="5"/>
```

```
    <voice>Alice</voice>
    Good morning.
    <voice>Bob</voice>
    Morning - did you see the report?
    <bdiv/>
    <minutes value="59"/><seconds value="58"/><msecs value="450"/>
    <hours value="1"/><minutes value="0"/><seconds value="1"/><msecs value="210"/>
    <frame><src uri="assets/frames/00-59-58.jpg"/></frame>
    Wrapping up as we pass the hour.
  </track>
</doclang>
```

<cover>

A representative image for a `<track>` as a whole — for example podcast artwork, a poster, or a title card. Appears once, before the first cue block. Same shape as `<frame>`: an optional element head followed by an optional `<src>`.

Allowed context

Can only appear inside a `<track>`, before the first `<bdiv>`.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Not allowed
Primary semantic elements	Allowed

<frame>

The still image of the recording at a `<track>` cue block's start time. Shaped like `<picture>`: an optional element head followed by an optional `<src>`.

Allowed context

Can only appear inside a `<track>` cue block.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Not allowed
Primary semantic elements	Allowed

<audio>

The recording over a `<track>` cue block's interval `[start, end]` — a temporal crop of the track's audio, as a `<picture>` crop corresponds to its page region. Requires the cue block to

carry an end time; the timestamps are authoritative and the clip is a best-effort fragment. Same shape as `<frame>`.

Allowed context

Can only appear inside a `<track>` cue block.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Not allowed
Primary semantic elements	Allowed

<voice>

Speaker attribution for a transcript turn within a `<track>` cue block. Handled like `<text>` (may carry its own element head and inline formatting).

Allowed context

Can only appear inside a `<track>` cue block.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

<chapter>

A chapter or section title for a `<track>`. It marks a chapter boundary at the start time of the cue block that carries it; the chapter runs until the next `<chapter>` (or `</track>`), and the region before the first `<chapter>` is unchaptered. Only the start time matters — the cue block's end time bounds its transcript, not the chapter. Chapters form a flat partition (no nesting, no overlap). Handled like `<text>` (may carry its own element head and inline formatting).

Allowed context

Can only appear inside a `<track>` cue block, after the start (and optional end) timestamp and before any `<frame>`.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Allowed
Raw text	Allowed
Primary semantic elements	Allowed

Property elements

Property elements are non-semantic elements that help define useful traits of a semantic element, forming the main building blocks of the element head. Property elements that can appear on the top level of the element head are called *primary*, while those that can only appear within other property elements are called *secondary*. The various property elements are further specified in the following subsections.

<label>

Optional part of the element head; serves for providing a detailed label for the respective element.

Allowed context

Can only be part of the element head of a semantic element.

Attributes

Attribute	Required / Optional	Allowed values	Description
`value`	Optional; default: "undefined"	Different value domains may be recommended per host element (not to be validated).	A label for concretely specifying a subclass type for the host element.

Allowed content types

None (empty element).

<thread>

Optional part of the element head; serves for establishing a logical document component. This can be useful for capturing fragmented components, e.g. spanning multiple bounding boxes (e.g. cross-column) or pages, or for defining anchors for cross references. To capture a fragmented component, we define separate instances of the respective element and use a `<thread>` with the same `thread_id` attribute for all of them

Allowed context

Can only be part of the element head of a semantic element.

Attributes

Attribute	Required / Optional	Allowed values	Description
`thread_id`	Required	Positive integer	The ID of the thread. All `<thread>`

	elements that share a given `thread_id` must be under the same host element type (e.g. all under `<text>`, not mixed `<text>` and `<picture>`).

Allowed content types

None (empty element).

<xref>

Optional part of the element head; serves for capturing an outgoing cross-reference from this component.

Allowed context

Can only be part of the element head of a semantic element.

Attributes

Attribute	Required / Optional	Allowed values	Description
`thread_id`	Required	Positive integer	The ID of the referenced thread. This must be defined by at least one `<thread>` in the DocLang instance.

Allowed content types

None (empty element).

<href>

Optional part of the element head; serves for capturing a URI referenced by this component.

Allowed context

Can only be part of the element head of a semantic element.

Attributes

Attribute	Required / Optional	Allowed values	Description
`uri`	Required	URI	The URI of the referenced resource.

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed

Raw text	Not allowed
Primary semantic elements	Not allowed

<description>

Optional part of the element head for capturing a derived textual account of what the host component is or what it shows. Unlike `<caption>`, `<description>` is meta-information and not an actual document component. For example, a picture description inferred by a model.

Allowed context

Can only be part of the element head of a semantic element.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<summary>

Optional part of the element head for capturing a derived textual distillation of what the host component conveys. Unlike `<caption>`, `<summary>` is meta-information and not part of the original document content.

Allowed context

Can only be part of the element head of a semantic element.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<custom>

Optional part of the element head; custom metadata, e.g. for application-specific purposes. See Recommendations for naming and namespacing guidance for custom vocabularies.

Allowed context

Can only be part of the element head of a semantic element.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Not allowed
Primary semantic elements	Not allowed

<location>

Optional part of the element head; coordinate system is the top-left corner of the page. When present, appears in sequence of 4 `<location>` instances, representing x0, y0, x1, y1. The following must then hold: `x0_norm<=x1_norm` and `y0_norm<=y1_norm`(i.e. first top-left point then bottom-right point), whereby `*_norm` is the respective coordinate normalized to its effective resolution.

Allowed context

Can only be part of the element head of a semantic element.

Attributes

Attribute	Required / Optional	Allowed values	Description
`resolution`	Optional; defaults to `default_resolution@width` or `default_resolution@height` depending on whether location refers to x or y, otherwise "512" if respective `default_resolution` not explicitly specified	Positive integer	Axis boundary (exclusive) for the respective `location@value`.
`value`	Required	Integer within [0, `location@resolution`)	Coordinate w.r.t. top-left corner.

Allowed content types

None (empty element).

<layer>

The conceptual layer of the host element in the DocLang instance.

Allowed context

Can only be part of the element head of a semantic element.

Attributes

Attribute	Required / Optional	Allowed values	Description
`value`	Optional; default: "body"	{"body", "background", "furniture"}	The layer value: "body" is for the main content, "background" is for

	watermarks and other background components, while "furniture" is the fallback for any supplementary components not contributing to the main content, such as navigation or decorations.

Allowed content types

None (empty element).

Payload elements

Payload elements are low-level elements that help define the effective content of another element.

<src>

The source of the element's media: an image in `<picture>`, `<frame>` and `<cover>`, or an audio clip in `<audio>`.

Allowed context

Can only be part of the picture-specific element body sequence of `<picture>`, or come right after the element head of `<frame>`, `<cover>` or `<audio>`.

Attributes

Attribute	Required / Optional	Allowed values	Description
`uri`	Required	URI	The source URI. May use a `data:` URI (RFC 2397) with base64-encoded payload, e.g. `uri="…"`. Relative URIs are allowed too, e.g. `uri="assets/chart.svg"`.

Allowed content types

None (empty element).

<tabular>

Structured tabular data. Uses the same cell model as `<table>`, but without an element head of its own.

Allowed context

Can only be part of the picture-specific element body sequence of `<picture class="chart">`.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Only on cell level, in case of virtual `<text>`
Raw text	Only on cell level, in case of virtual `<text>`
Primary semantic elements	Allowed

<checkbox>

Empty element representing checkbox selection state.

Allowed context

Any context that allows raw text content.

Attributes

Attribute	Required / Optional	Allowed values	Description
`class`	Optional; default: "unselected"	{"unselected", "selected"}	The checkbox type.

Allowed content types

None (empty element).

<content>

Whitespace-preserving text container. Retains all whitespace within the element (equivalent to `xml:space="preserve"`), enabling use in whitespace-sensitive scenarios such as code blocks. XML special characters may alternatively be conveyed via CDATA.

Allowed context

Any context that allows raw text content.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<hours>

Hours component of a `<track>` cue timestamp. Optional within a timestamp run (defaults to 0).

Allowed context

Can only be part of a timestamp run in a `<track>` cue block.

Attributes

Attribute	Required / Optional	Allowed values	Description
`value`	Required	Non-negative integer	Hours component of a track cue timestamp.

Allowed content types

None (empty element).

<minutes>

Minutes component of a `<track>` cue timestamp. Optional within a timestamp run (defaults to 0).

Allowed context

Can only be part of a timestamp run in a `<track>` cue block.

Attributes

Attribute	Required / Optional	Allowed values	Description
`value`	Required	Integer within [0, 59]	Minutes component of a track cue timestamp.

Allowed content types

None (empty element).

<seconds>

Seconds component of a `<track>` cue timestamp.

Allowed context

Can only be part of a timestamp run in a `<track>` cue block.

Attributes

Attribute	Required / Optional	Allowed values	Description
`value`	Required	Integer within [0, 59]	Seconds component of a track cue timestamp.

Allowed content types

None (empty element).

<msecs>

Milliseconds component of a `<track>` cue timestamp. Optional within a timestamp run (defaults to 0).

Allowed context

Can only be part of a timestamp run in a `<track>` cue block.

Attributes

Attribute	Required / Optional	Allowed values	Description
`value`	Required	Integer within [0, 999]	Milliseconds component of a track cue timestamp.

Allowed content types

None (empty element).

Formatting elements

Formatting elements modify the styling and presentation within semantic or other formatting elements.

<bold>

Indicates bold formatting.

Allowed context

Any context that allows raw text content.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<italic>

Indicates italic formatting.

Allowed context

Any context that allows raw text content.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<underline>

Indicates underlined formatting.

Allowed context

Any context that allows raw text content.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<strikethrough>

Indicates struck-through formatting.

Allowed context

Any context that allows raw text content.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<superscript>

Indicates superscript formatting.

Allowed context

Any context that allows raw text content.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<subscript>

Indicates subscript formatting.

Allowed context

Any context that allows raw text content.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<handwriting>

Indicates handwritten text.

Allowed context

Any context that allows raw text content.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

<rtl>

Indicates right-to-left direction.

Allowed context

Any context that allows raw text content.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Allowed
Primary semantic elements	Not allowed

Structural elements

Structural elements define boundaries within explicitly structured components like tables and lists.

<fcel>

Indicates the beginning of a full / regular cell.

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<ecel>

Indicates the beginning of an empty cell.

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<ched>

Indicates the beginning of a column header cell.

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<rhed>

Indicates the beginning of a row header cell.

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<corn>

Indicates the beginning of a corner cell, typically the top-left header intersection.

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<srow>

Indicates the beginning of a section row header.

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<lcel>

Left-merge extension token; extends the previous cell horizontally (colspan continuation).

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<ucel>

Upward-merge extension token; extends the cell above vertically (rowspan continuation)

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<xcel>

Cross/combined span extension token; used where both horizontal and vertical spanning intersect.

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<nl>

Denotes the end of a table row.

Allowed context

Can only be child of `<table>`, `<index>`, or `<tabular>`

Attributes

None

Allowed content types

None (empty element).

<ldiv>

Indicates the beginning of a list item. It can either be empty or contain a `<marker>`.

Allowed context

Can only be child of `<list>`.

Attributes

None

Allowed content types

Content type	Allowed / Not allowed
Element head	Not allowed
Raw text	Not allowed
Primary semantic elements	Only `<marker>`

<bdiv>

Indicates the beginning of a `<track>` cue block.

Allowed context

Can only be a child of `<track>`.

Attributes

None

Allowed content types

None (empty element).

Document head elements

This category comprises the document-level metadata elements that are the building blocks of `<head>`.

<default_resolution>

Defines the default resolution for the DocLang instance.

Allowed context

Can only be child of `<head>`.

Attributes

Attribute	Required / Optional	Allowed values	Description
`width`	Optional; default "512"	Non-negative integer	Default document width in pixels.
`height`	Optional; default "512"	Non-negative integer	Default document height in pixels.

Allowed content types

None (empty element).

Annex B (normative): DocLang archive format

A **DocLang archive** is a ZIP file using the Open Packaging Conventions (OPC) container model. Recommended extension: **`.dclx`**.

An archive is a **package** of **parts** (files in the ZIP). Each part has a path (e.g. `/document.xml`) and a content type declared in `[Content_Types].xml`. The package root relationship file `_rels/.rels` identifies the main DocLang instance part.

Layout

```
[Content_Types].xml    # required — OPC content types
_rels/
  .rels                # required — package → main document
document.xml           # required — valid DocLang instance
pages/                 # optional — page images, e.g. 1.png, 2.png
audio/                 # optional — whole-track audio, e.g. 2.mp3, 4.ogg
video/                 # optional — whole-track video, e.g. 3.mp4, 4.mkv
assets/                # optional — files referenced from markup
```

[Content_Types].xml (required)

Declares content types for package parts. MUST include an `<Override>` for `/document.xml` with content type `application/vnd.doclang.document+xml`, and a

`<Default>` (or per-part `<Override>`) for **every** extension present in the package. SHOULD declare defaults for common image extensions used under `pages/` (`png`, `jpg`, `jpeg`, `webp`), for audio/video extensions used under `audio/` / `video/` (e.g. `mp3` → `audio/mpeg`, `m4a` → `audio/mp4`, `ogg`/`opus` → `audio/ogg`, `wav` → `audio/wav`, `flac` → `audio/flac`, `mp4` → `video/mp4`, `webm` → `video/webm`, `mkv` → `video/x-matroska`, `mov` → `video/quicktime`), and for `.rels` parts. This media type may be registered with the Internet Assigned Numbers Authority (IANA) in a future revision; conformance does not depend on registration.

Example:

```
<?xml version="1.0" encoding="UTF-8"?>
<Types xmlns="http://schemas.openxmlformats.org/package/2006/content-types">
  <Default Extension="rels" ContentType="application/vnd.openxmlformats-
package.relationships+xml"/>
  <Default Extension="png" ContentType="image/png"/>
  <Default Extension="mp3" ContentType="audio/mpeg"/>
  <Default Extension="mp4" ContentType="video/mp4"/>
  <Override PartName="/document.xml"
ContentType="application/vnd.doclang.document+xml"/>
</Types>
```

_rels/.rels (required)

Package-level relationships. MUST contain exactly one relationship of type `http://doclang.ai/ns/package/2026/relationships/document` targeting `document.xml`.

Example:

```
<?xml version="1.0" encoding="UTF-8"?>
<Relationships xmlns="http://schemas.openxmlformats.org/package/2006/relationships">
  <Relationship Id="rId1"
    Type="http://doclang.ai/ns/package/2026/relationships/document"
    Target="document.xml"/>
</Relationships>
```

document.xml (required)

The main part: a valid DocLang instance. Top-level `<page_break/>` elements separate pages; `<location>` coordinates are page-relative.

pages/ (optional)

Optional raster images for review. Name files `{N}.{png|jpg|jpeg|webp}` with 1-based `N`. Gaps are allowed.

audio/ (optional)

video/ (optional)

Optional whole-track media. Name files `{N}.{ext}` where `N` is the 1-based position of the corresponding `<track>` in document reading order (counting every `<track>`, including nested ones). A track has a whole-track audio recording when `audio/{N}.*` is present, and

a whole-track video recording when `video/{N}.*` is present — either, both, or neither. At most one file per track per modality. Gaps are allowed; `N` MUST NOT exceed the number of `<track>` elements in the DocLang instance.

assets/ (optional)

Optional payload referenced by relative URIs in markup (e.g. `<src uri="assets/chart.svg"/>`). URIs resolve from the archive root.

Page alignment

Review tools split markup (excluding `<head>`) on `<page_break/>`; segment *N* corresponds to page *N*. Page count is determined by markup: the number of `<page_break/>` elements plus one. Page images are optional and individual gaps are allowed, but page files must not exceed this count (e.g. two `<page_break/>` elements define three pages, so `4.png` is out of bounds).

Track alignment

The *N*-th `<track>` in the DocLang instance aligns with `audio/{N}.*` and/or `video/{N}.*` when present. A cue block's timestamps are then playback offsets into that file, measured from its start (`00:00:00.000`). Per-cue `<frame>` and `<audio>` fragments are independent references (via `<src>`) and are not derived from the whole-track files. A track with neither an `audio/` nor a `video/` entry is simply a transcript whose timestamps refer to a recording the archive does not carry.

Annex C (informative): Recommendations

This appendix is informative and does not define conformance requirements.

Recommended labels

Pictures

For the `label@value` of `<picture>` elements, we recommend using the values defined below:

Context	Recommended values
`<picture class="chart">`	`bar_chart`, `box_plot`, `flow_chart`, `line_chart`, `pie_chart`, `scatter_plot`
else, i.e. `<picture class="undefined">` (default)	`full_page_image`, `page_thumbnail`, `photograph`, `chemistry_structure`, `bar_code`, `icon`, `logo`, `qr_code`, `signature`, `stamp`, `engineering_drawing`, `screenshot_from_computer`, `screenshot_from_manual`, `geographical_map`, `topographical_map`, `calendar`, `crossword_puzzle`, `music`

Additional special cases:

- `other`: use when the picture was examined but does not fit any of the recommended values from above (e.g. a chart that is not bar/box/flow/line/pie/scatter).
- `undefined`: use when classification has not been performed (default `label@value`).

Note: `picture@class="undefined"` (default picture type) and `label@value="undefined"` (unclassified subclass) are independent.

Code

For the `label@value` of `<code>` elements, we recommend using the values defined below:

Context	Recommended values
`<code>`	Linguist v9.5.0 language keys (e.g. `Python`)

Additional special cases:

- `other`: use when the code was examined but does not match any of the recommended values from above.
- `undefined`: use when classification has not been performed (default `label@value`).

Custom vocabulary naming and namespacing

Content inside `<custom>` is implementation-defined and not governed by this document. To improve interoperability and reduce naming collisions, the following recommendations apply:

- Producers of shared custom vocabularies SHOULD use formal XML namespaces with stable namespace URIs.
- Processors SHOULD treat the namespace URI as the namespace identifier and MUST NOT assume semantic meaning from prefix names alone.
- For private or local use where formal namespaces are not used, producers SHOULD use collision-resistant element names prefixed with a stable organization or project identifier (e.g. `acme_`).
- Producers SHOULD avoid ambiguous generic names such as `item`, `meta`, or `data` unless these are clearly scoped by namespace or equivalent prefixing.
- Producers SHOULD document custom vocabularies (intended meaning, value domains, and versioning policy) when documents are shared across tools or organizations.

Token vocabulary

Below is a list of tokens that can be used for a DocLang-compliant tokenizer.

The token vocabulary trades off size and inference cost:

- For frequent elements such as `<location>` (often paired with semantic elements), the vocabulary defines one token per concrete instance — e.g. `<location value="0"/>`, `<location value="1"/>`, ..., `<location value="511"/>` — to reduce inference overhead.

- For rare elements such as `<thread>`, the vocabulary uses separate tokens — e.g. `<thread thread_id="`, the id, `"/>` — to limit vocabulary growth at the cost of slightly more inference work.

Token	Description
`<doclang>`	`doclang` start
`</doclang>`	`doclang` end
`<page_break/>`	`page_break`
`"/>`	end of self-closing element with attributes
`<text>`	`text` start
`</text>`	`text` end
`<heading>`	level-1 `heading` start
`<heading level="2">`	level-2 `heading` start
`<heading level="3">`	level-3 `heading` start
`<heading level="4">`	level-4 `heading` start
`<heading level="5">`	level-5 `heading` start
`<heading level="6">`	level-6 `heading` start
`</heading>`	`heading` end
`<footnote>`	`footnote` start
`</footnote>`	`footnote` end
`<page_header>`	`page_header` start
`</page_header>`	`page_header` end
`<page_footer>`	`page_footer` start
`</page_footer>`	`page_footer` end
`<field_region>`	`field_region` start
`</field_region>`	`field_region` end
`<list>`	unordered `list` start
`<list class="ordered">`	ordered `list` start
`</list>`	`list` end
`<table>`	`table` start
`</table>`	`table` end
`<index>`	`index` start
`</index>`	`index` end
`<formula>`	`formula` start
`</formula>`	`formula` end
`<code><label value="`	start of `code` with `label`
`</code>`	`code` end
`<picture><label value="`	start of `picture` with `label`
`<picture class="chart"><label value="`	start of chart `picture` with `label`
`</picture>`	`picture` end
`<marker>`	`marker` start
`</marker>`	`marker` end
`<group>`	`group` start
`</group>`	`group` end

<field_heading>	level-1 field_heading start
<field_heading level="2">	level-2 field_heading start
<field_heading level="3">	level-3 field_heading start
<field_heading level="4">	level-4 field_heading start
<field_heading level="5">	level-5 field_heading start
<field_heading level="6">	level-6 field_heading start
</field_heading>	field_heading end
<field_item>	field_item start
</field_item>	field_item end
<key>	key start
</key>	key end
<value>	read-only value start
<value class="fillable">	fillable value start
</value>	value end
<hint>	hint start
</hint>	hint end
<caption>	caption start
</caption>	caption end
<description>	description start
</description>	description end
<summary>	summary start
</summary>	summary end
<thread thread_id="	thread with thread_id attribute start
<xref thread_id="	xref with thread_id attribute start
<href uri="	href with uri attribute start
<custom>	custom start
</custom>	custom end
<layer value="	layer with value attribute start
<src uri="	src with uri attribute start
<tabular>	tabular start
</tabular>	tabular end
<checkbox class="unselected"/>	unselected checkbox
<checkbox class="selected"/>	selected checkbox
<content>	content start
</content>	content end
<![CDATA[	CDATA section start
]]>	CDATA section end
<bold>	bold start
</bold>	bold end
<italic>	italic start
</italic>	italic end
<underline>	underline start
</underline>	underline end
<strikethrough>	strikethrough start
</strikethrough>	strikethrough end

<superscript>	superscript start
</superscript>	superscript end
<subscript>	subscript start
</subscript>	subscript end
<handwriting>	handwriting start
</handwriting>	handwriting end
<rtl>	rtl start
</rtl>	rtl end
<fcel/>	fcel
<ecel/>	ecel
<ched/>	ched
<rhed/>	rhed
<corn/>	corn
<srow/>	srow
<lcel/>	lcel
<ucel/>	ucel
<xcel/>	xcel
<nl/>	nl
<ldiv/>	ldiv
<ldiv><marker>	start of ldiv with marker
</marker></ldiv>	end of ldiv with marker
<track>	track start
</track>	track end
<bdiv/>	bdiv cue-block delimiter
<cover>	cover start
</cover>	cover end
<frame>	frame start
</frame>	frame end
<audio>	audio start
</audio>	audio end
<voice>	voice start
</voice>	voice end
<chapter>	chapter start
</chapter>	chapter end
<hours value="	hours with value attribute start (out-of-range values)
<minutes value="	minutes with value attribute start (out-of-range values)
<seconds value="	seconds with value attribute start (out-of-range values)
<msecs value="	msecs with value attribute start (out-of-range values)
<location value="0"/>, **<location value="1"/>**, ..., **<location value="511"/>**	location tokens with values from 0 to 511
<hours value="0"/>, **<hours**	hours tokens with values from 0 to 23

`value="1"/>, ..., <hours value="23"/>`	
`<minutes value="0"/>, <minutes value="1"/>, ..., <minutes value="59"/>`	`minutes` tokens with values from 0 to 59
`<seconds value="0"/>, <seconds value="1"/>, ..., <seconds value="59"/>`	`seconds` tokens with values from 0 to 59
`<msecs value="0"/>, <msecs value="10"/>, ..., <msecs value="990"/>`	`msecs` tokens with values in multiples of 10 from 0 to 990

Annex D (informative): Future extensions

These features are considered for future versions of Doclang.

Horizontal threading

Horizontal threading enables linking related content across the horizontal axis in page or table layouts. The `h_thread` element supports this by explicitly connecting content (like table rows or columns) that spans multiple pages, ensuring these threads remain structured and traceable.

Explicitly wrapped repetitive items

Repetitive structures currently rely on a bare delimiter plus unwrapped content: `<ldiv/>` for list items, cell tokens for table cells, and `<bdiv/>` for track cue blocks. A future revision may allow each item to alternatively be written as an explicit wrapper element carrying its own element head, in a way that stays backward compatible with the delimiter form and remains consistent across all three modalities.

Multi-part track media

The archive format carries at most one whole-track file per `<track>` per modality (`audio/{N}.*`, `video/{N}.*`). A future revision may add a backward-compatible form for media split into ordered segments (e.g. an `audio/{N}/` directory plus a small manifest giving each segment's start offset), for very long recordings or capture that was paused and resumed.

Metadata

<head>

Document-level metadata is contained in the optional `<head>` element.

Below we list the reserved core metadata elements to be used within `<head>`:

- `title`
- each `author` element can optionally begin with one or more `affiliation` elements
- `date`
- `page_size`, the actual page size. An element without the `page_no` attribute defines the default size for all pages, when `page_no` is specified it is counted from 1.

- `language`, Identifies the (human) language of the document, e.g., English, German, French, Spanish, Japanese. The content MUST be an ISO 639-3 language identifier. Optional attributes: `classifier` (the tool/method used, e.g., fastText) and `score` (confidence in [0, 1]). Multiple `language` entries MAY be provided.
- `generated_by`, upstream pipeline information, e.g. vision language model (VLM) ID
- `topic`, topic that the document is most likely to fall in such as Science and Technology, Legal, etc. The topics should preferrably come from some taxonomy. Classifier defines the classifier used for classifying into the given topic and score is the confidence score of classifier and 0<=Scores<=1. This can be one or more.
- `summary`, a summary of the document (document-level; distinct from element-head `<summary>` on individual components)
- `document_hash`, Hash of the source document, whereas hash_function defines the algorithm used to compute the hash, e.g., SHA2. This can be one or more.

Here is an example:

```
<doclang>
  <head>
    <!-- reserved elements -->
    <title>My Company's Annual Report</title>
    <author_info>
      <author>Author 1 Name</author>
      <author>
        <affiliation>Affiliation A</affiliation>
        <affiliation>Affiliation B</affiliation>
        Author 2 Name
      </author>
    </author_info>
    <date>2024-01-01</date>
    <language classifier="fastText" score="0.7">eng</language>
    <language classifier="fastText" score="0.2">spa</language>
    <topic topic_taxonomy="taxonomy" score="0.5">Technology</topic>
    <topic topic_taxonomy="taxonomy" score="0.5">Math</topic>
    <document_hash
hash_function="sha256sum">75f2db0c6124527bf6dd48440f95fc864a5108d28517633f937923a7d8199
185</document_hash>
    <summary>This is a summary of the document</summary>
    <generated_by>example_vlm_org/example_vlm_name</generated_by>
    <default_resolution width="512" height="512"/>
    <page_size width="612" height="792"/>
    <page_size page_no="4" width="792" height="612"/>

      <!-- examples of custom elements -->
    <my_company_hap_filter_hate>0.1</my_company_hap_filter_hate>
    <my_company_hap_filter_abuse>0.1</my_company_hap_filter_abuse>
    <my_company_hap_filter_profanity>0.1</my_company_hap_filter_profanity>
  </head>
  <!-- document content -->
</doclang>
```

Governance and compliance metadata

In addition to the core metadata elements, publishers can optionally provide metadata pertaining to governance and compliance. These elements allow the communication of acceptable use, policy, licensing, contact information and compliance requirements.

Governance and compliance metadata MUST be expressed at the document level inside `<head>` and MAY be overridden at component level for finer-grained control.

Standards reference and interpretation

The standards referenced throughout this Governance and Compliance section are provided to support alignment, interpretation, and interoperability. They are **informative**, not normative, unless explicitly stated otherwise.

These references are intended to:

- Clarify regulatory or industry concepts reflected by the metadata elements
- Assist implementers in mapping DocLang governance signals to existing compliance programs
- Avoid re-defining legal or regulatory obligations within this document

DocLang governance metadata does **not** claim conformance to any listed framework by itself. Rather, it provides structured, machine-readable signals that downstream systems MAY use to support compliance, risk management, and audit workflows.

The following standards and regulations are commonly referenced:

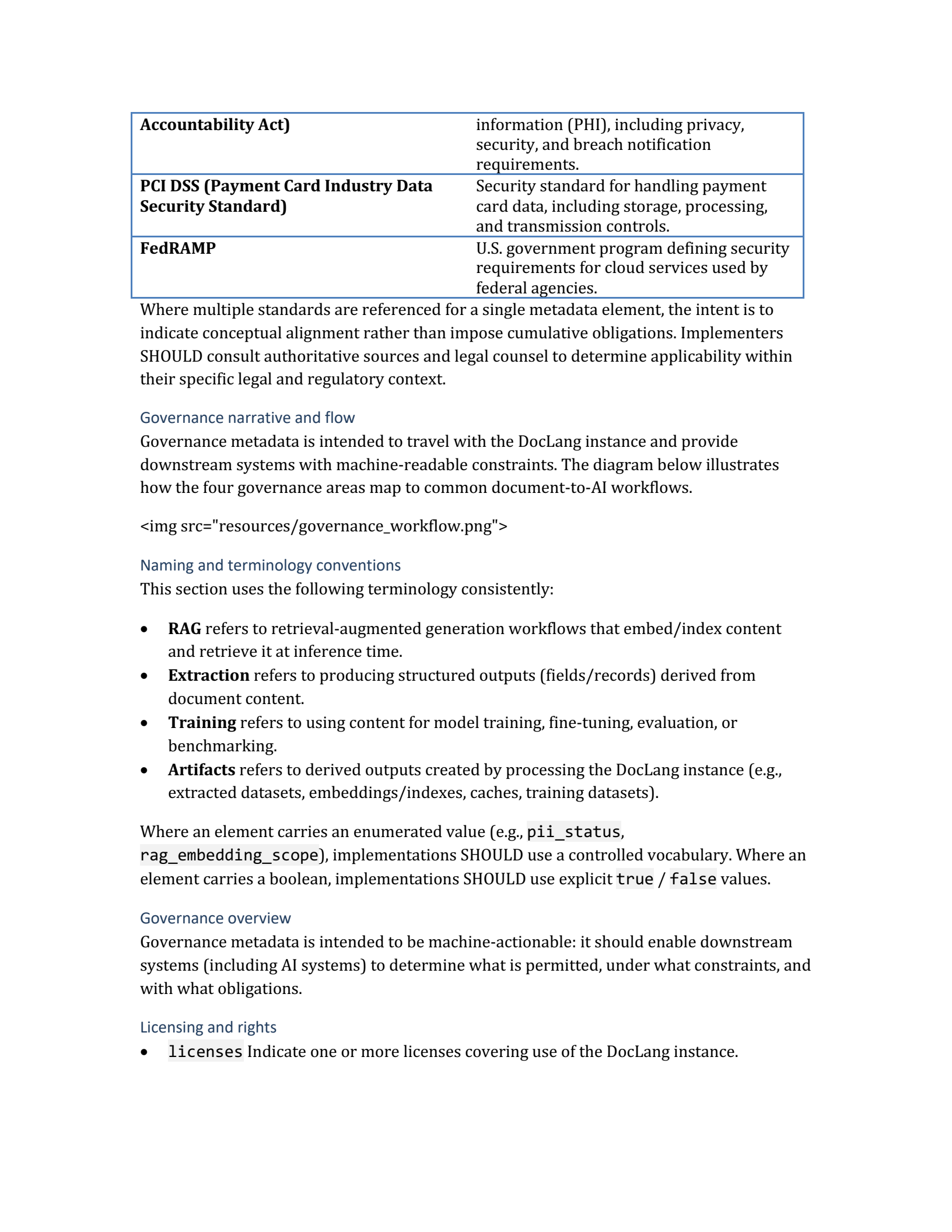
Standard / Regulation	Scope and relevance
GDPR (EU General Data Protection Regulation)	Governs processing of personal data, including lawful basis, purpose limitation, data minimization, retention, data subject rights, and cross-border transfer obligations.
EU AI Act	Establishes obligations related to AI system risk categories, data governance, transparency, and human oversight for AI systems placed on or used within the EU.
ISO/IEC 27001	International standard for information security management systems (ISMS), covering access control, asset management, logging, and security controls.
ISO/IEC 27701	Extension to ISO/IEC 27001 focused on privacy information management (PIMS), including controller/processor roles and personal data handling.
ISO/IEC 23894	Guidance on AI risk management, including lifecycle governance, human oversight, and accountability for AI systems.
HIPAA (Health Insurance Portability and	U.S. regulation governing protected health

Accountability Act)	information (PHI), including privacy, security, and breach notification requirements.
PCI DSS (Payment Card Industry Data Security Standard)	Security standard for handling payment card data, including storage, processing, and transmission controls.
FedRAMP	U.S. government program defining security requirements for cloud services used by federal agencies.

Where multiple standards are referenced for a single metadata element, the intent is to indicate conceptual alignment rather than impose cumulative obligations. Implementers SHOULD consult authoritative sources and legal counsel to determine applicability within their specific legal and regulatory context.

Governance narrative and flow

Governance metadata is intended to travel with the DocLang instance and provide downstream systems with machine-readable constraints. The diagram below illustrates how the four governance areas map to common document-to-AI workflows.



Naming and terminology conventions

This section uses the following terminology consistently:

- **RAG** refers to retrieval-augmented generation workflows that embed/index content and retrieve it at inference time.
- **Extraction** refers to producing structured outputs (fields/records) derived from document content.
- **Training** refers to using content for model training, fine-tuning, evaluation, or benchmarking.
- **Artifacts** refers to derived outputs created by processing the DocLang instance (e.g., extracted datasets, embeddings/indexes, caches, training datasets).

Where an element carries an enumerated value (e.g., `pii_status`, `rag_embedding_scope`), implementations SHOULD use a controlled vocabulary. Where an element carries a boolean, implementations SHOULD use explicit `true` / `false` values.

Governance overview

Governance metadata is intended to be machine-actionable: it should enable downstream systems (including AI systems) to determine what is permitted, under what constraints, and with what obligations.

Licensing and rights

- `licenses` Indicate one or more licenses covering use of the DocLang instance.

Data classification and privacy posture

- `data_classification` One or more data classifications can be given for the document content.

In general, data classification is not globally standardized. Organizations usually define a classification system suitable for their respective mission. These elements allow an organization to classify document sensitivity in their own terms.

Acceptable use and purpose limitation

- `acceptable_use` Organizations may express acceptable use cases for the provided document data.

Stewardship and contact

- `stewardship` Provides the name of a person and/or organization with governance responsibility at the owning organization.

Access control policy

- `access_policy` Provides the ability to express access policy as well as enumerate roles allowed to access the data.

Similar to data classification, there are no standards specifying role semantics. This element allows organizations to describe access policy and roles in their own terms.

Retention and deletion

- `retention_policy` Allows organizations to state retention objectives.

Compliance frameworks

- `compliance_requirements` States the compliance frameworks (regulatory or industrial) governing the lifecycle and use of the documents.

Privacy and personally identifiable information (PII) controls

This subsection defines governance signals related to personal data detection, sensitivity, and permitted handling. Implementations SHOULD use these elements to drive privacy-aware processing (e.g., redaction, restricted access, minimization).

The following optional elements MAY be provided to describe personal data presence, sensitivity, permitted processing, and privacy-related obligations. Unless otherwise required by an implementation, these elements are intended to be expressed at the document level inside `<head>`.

Element	Purpose	Standards alignment (non-exhaustive)
`pii_status`	Indicates whether the DocLang instance contains PII.	ISO 27701; ISO 27001 A.5/A.8; GDPR Art. 4(1),

		5(1)
pii_sensitivity_level	Classifies the sensitivity level of detected PII.	ISO 27701; GDPR Art. 9–10
pii_source_type	Identifies the origin/source of PII (e.g., provided by user, derived, third-party).	GDPR Art. 13–14
controller_processor_role	Defines the organizational role for processing (controller/processor or equivalent).	GDPR Art. 24–28
pii_processing_purpose	Specifies the purpose of processing (purpose limitation).	GDPR Art. 5(1)(b), Art. 6
pii_lawful_basis	Records lawful basis for processing.	GDPR Art. 6(1)
special_category_condition	Condition for processing special-category data (if applicable).	GDPR Art. 9(2)
pii_minimisation_status	Indicates whether minimisation has been applied (data minimisation / privacy by design).	GDPR Art. 5(1)(c), Art. 25
pii_transformation_level	Indicates transformation applied to PII (e.g., redacted, masked, pseudonymized).	GDPR Recital 26; GDPR Art. 4(5)
reidentification_risk	Expresses assessed risk of re-identification where transformations are used.	ISO 27701; GDPR Recital 26
access_control_level	Required access tier/controls for handling this DocLang instance.	ISO 27001 A.9; GDPR Art. 32
ai_use_restriction	Indicates allowed AI uses or prohibitions for the content of this DocLang instance.	GDPR Art. 5(1)(b–c)
cross_border_transfer_status	Indicates whether cross-border transfers occur/are allowed.	GDPR Art. 44–49
transfer_mechanism	Indicates transfer mechanism where applicable (e.g., adequacy, SCCs).	GDPR Art. 45–47
retention_category	Indicates retention category for personal data contained in the DocLang instance.	GDPR Art. 5(1)(e)
dsr_impact_flag	Signals potential impact on the handling of data subject rights (DSR).	GDPR Art. 12–23
dpia_required	Indicates whether a data	GDPR Art. 35–36

	protection impact assessment (DPIA) is required for intended processing.	
`children_pii_present`	Flags whether children's data is present.	GDPR Art. 8
`automated_decisioning_relevance`	Indicates whether automated decision-making/profiling obligations apply.	GDPR Art. 22
`logging_monitoring_enabled`	Indicates whether logging/monitoring is enabled for access and processing (accountability/auditability).	ISO 27001 A.12/A.16; GDPR Art. 5(2)

Implementations SHOULD define controlled vocabularies (and, where applicable, boolean conventions) for these elements. If an organization already has established internal taxonomies for classification, purpose, lawful basis, access tiers, or transfer mechanisms, those SHOULD be used consistently.

Data extraction controls

This subsection defines governance signals that constrain automated extraction, transformation, and downstream use of extracted fields. Implementations SHOULD use these elements to ensure purpose limitation and auditability of extraction.

The following optional elements MAY be provided to express constraints and obligations related to automated or manual data extraction from the DocLang instance. These elements are intended to guide downstream systems that perform field extraction, transformation, enrichment, or export. Unless otherwise required by an implementation, these elements SHOULD be expressed at the document level inside `<head>`, and MAY be overridden at component level for finer-grained control.

Element	Purpose	Standards alignment (non-exhaustive)
`extraction_permitted`	Indicates whether automated data extraction is permitted at all.	GDPR Art. 5(1)(a,b); ISO 27701
`extraction_scope`	Defines which parts or components of the DocLang instance may be extracted (e.g., full document, tables only, specific sections).	GDPR Art. 5(1)(b,c)
`extraction_purpose`	Specifies the allowed purpose(s) for	GDPR Art. 5(1)(b); ISO 27701

	extracted data.	
`extraction_granularity`	Indicates permitted level of granularity (e.g., aggregate only, field-level, record-level).	GDPR Art. 5(1)(c)
`pii_extraction_allowed`	Indicates whether PII may be included in extracted outputs.	GDPR Art. 6; ISO 27701
`sensitive_data_extraction_allowed`	Indicates whether special-category or sensitive data may be extracted.	GDPR Art. 9–10
`extraction_transformation_required`	Specifies required transformations during extraction (e.g., redaction, masking, pseudonymization).	GDPR Art. 25; Recital 26
`extraction_output_constraints`	Constrains allowed output formats or destinations for extracted data.	ISO 27001 A.8; GDPR Art. 32
`downstream_sharing_permitted`	Indicates whether extracted data may be shared with downstream systems or third parties.	GDPR Art. 5(1)(a,b); Art. 28
`downstream_usage_restrictions`	Specifies restrictions on how extracted data may be used downstream.	GDPR Art. 5(1)(b)
`extraction_audit_required`	Indicates whether extraction activities must be logged and auditable.	GDPR Art. 5(2); ISO 27001 A.12
`extraction_audit_retention`	Specifies retention period for extraction audit logs.	GDPR Art. 5(1)(e)
`human_in_the_loop_required`	Indicates whether human review/approval is required before or after extraction.	ISO 23894; ISO 27701
`automated_decisioning_dependency`	Indicates whether extracted data feeds automated decision-making systems.	GDPR Art. 22

Implementations SHOULD define controlled vocabularies for scope, purpose, granularity, transformations, and output constraints. Where extraction interacts with PII, these

elements SHOULD be interpreted in conjunction with the Privacy and PII controls defined above.

RAG and retrieval controls

This subsection defines governance signals that constrain whether and how document content may be embedded, indexed, chunked, retrieved, and presented to models during retrieval-augmented generation. Implementations SHOULD use these elements to control exposure, leakage risk, and attribution requirements.

The following optional elements MAY be provided to govern whether and how document content may be embedded, indexed, retrieved, and surfaced to models or users as part of retrieval-augmented generation (RAG) workflows. These elements are intended to control exposure risk, attribution, and downstream use of retrieved content. Unless otherwise required by an implementation, these elements SHOULD be expressed at the document level inside `<head>`, and MAY be overridden at component level for finer-grained control.

Element	Purpose	Standards alignment (non-exhaustive)
`rag_permitted`	Indicates whether the DocLang instance may be used in RAG workflows at all.	GDPR Art. 5(1)(a,b); ISO 27701
`rag_indexing_allowed`	Indicates whether the document content may be indexed or embedded for retrieval.	GDPR Art. 5(1)(b,c); ISO 27001 A.8
`rag_embedding_scope`	Defines which parts or components of the DocLang instance may be embedded (e.g., full document, summaries only, specific sections).	GDPR Art. 5(1)(b,c)
`rag_chunking_constraints`	Specifies constraints on chunking strategy (e.g., max size, boundaries, semantic-only).	ISO 23894; privacy-by-design principles
`rag_query_restrictions`	Defines restrictions on the types of queries that may retrieve this content.	GDPR Art. 5(1)(b)
`rag_output_attribution_required`	Indicates whether attribution or citation is required when content is retrieved	ISO 27001 A.18; copyright best practice

	or surfaced.	
`rag_output_transformation_required`	Specifies required transformations on retrieved content (e.g., summarization, redaction).	GDPR Art. 25; Recital 26
`rag_pii_exposure_allowed`	Indicates whether retrieved content may expose PII.	GDPR Art. 6; Art. 32
`rag_sensitive_data_exposure_allowed`	Indicates whether special-category or sensitive data may be exposed via retrieval.	GDPR Art. 9–10
`rag_downstream_sharing_permitted`	Indicates whether retrieved content may be shared beyond the immediate RAG response.	GDPR Art. 5(1)(a,b); Art. 28
`rag_caching_allowed`	Indicates whether retrieved content may be cached for performance or reuse.	ISO 27001 A.8; GDPR Art. 5(1)(e)
`rag_cache_retention`	Specifies retention period for cached embeddings or retrieved content.	GDPR Art. 5(1)(e)
`rag_audit_required`	Indicates whether retrieval events must be logged and auditable.	GDPR Art. 5(2); ISO 27001 A.12
`rag_audit_retention`	Specifies retention period for RAG access and retrieval logs.	GDPR Art. 5(1)(e)
`rag_model_scope`	Restricts which models or model classes may access this content via RAG.	ISO 23894; internal governance

Implementations SHOULD define controlled vocabularies for embedding scope, chunking constraints, query restrictions, and model scope. Where RAG interacts with PII or sensitive data, these elements SHOULD be interpreted in conjunction with the Privacy and PII controls defined above.

Document training controls

This subsection defines governance signals that constrain whether and how document content may be used for training, fine-tuning, or evaluation of models. Implementations

SHOULD use these elements to ensure licensing compliance, privacy protection, provenance tracking, and alignment with regulatory and contractual obligations.

The following optional elements MAY be provided to govern whether and how document content may be used for model training, fine-tuning, evaluation, or benchmarking. These elements are intended to ensure licensing compliance, privacy protection, provenance tracking, and alignment with regulatory and contractual obligations. Unless otherwise required by an implementation, these elements SHOULD be expressed at the document level inside `<head>`, and MAY be overridden at component level for finer-grained control.

Element	Purpose	Standards alignment (non-exhaustive)
`training_permitted`	Indicates whether the DocLang instance may be used for any form of model training or fine-tuning.	GDPR Art. 5(1)(a,b); ISO 27701
`training_scope`	Defines which parts or components of the DocLang instance may be used for training (e.g., full document, summaries only, specific sections).	GDPR Art. 5(1)(b,c)
`training_purpose`	Specifies the intended purpose of training (e.g., general models, domain-specific models, evaluation only).	GDPR Art. 5(1)(b)
`training_model_type`	Restricts the types or classes of models that may be trained using this content.	ISO 23894; internal governance
`training_data_retention`	Specifies retention period for training datasets derived from this DocLang instance.	GDPR Art. 5(1)(e)
`training_dataset_reuse_allowed`	Indicates whether derived training datasets may be reused beyond the initial training purpose.	GDPR Art. 5(1)(b)

`training_derivative_sharing_permitted`	Indicates whether trained models or derivatives may be shared with third parties.	GDPR Art. 28; licensing obligations
`training_pii_included`	Indicates whether training data may include PII.	GDPR Art. 6; ISO 27701
`training_sensitive_data_included`	Indicates whether special-category or sensitive data may be included in training.	GDPR Art. 9–10
`training_transformation_required`	Specifies required transformations prior to training (e.g., anonymization, pseudonymization).	GDPR Art. 25; Recital 26
`training_provenance_required`	Indicates whether provenance metadata must be retained for training records.	ISO 27001 A.12; AI accountability best practice
`training_audit_required`	Indicates whether training usage must be logged and auditable.	GDPR Art. 5(2); ISO 27001 A.12
`training_audit_retention`	Specifies retention period for training-related audit logs.	GDPR Art. 5(1)(e)
`model_output_usage_constraints`	Specifies constraints on use of models trained on this content (e.g., internal-only, non-commercial).	Licensing and intellectual property (IP) best practice
`right_to_be_forgotten_applicability`	Indicates whether erasure obligations apply to trained models or datasets.	GDPR Art. 17; emerging AI guidance

Implementations SHOULD define controlled vocabularies for training scope, purpose, model type, and transformation requirements. Where training involves personal or sensitive data, these elements SHOULD be interpreted in conjunction with the Privacy and PII controls defined above.

Minimal and full governance profiles

Implementations may adopt either a minimal governance profile (recommended baseline) or a full governance profile (richer control surface). These profiles are informative and provided to encourage consistent adoption.

Minimal profile (recommended baseline)

Area	Recommended elements
Licensing & compliance	`licenses`, `compliance_requirements`
Classification & access	`data_classification`, `access_policy` (or `access_control_level` where used), `retention_policy`
PII	`pii_status`, `pii_sensitivity_level` (when `pii_status` is `present`)
Extraction	`extraction_permitted`, `pii_extraction_allowed`
RAG	`rag_permitted`, `rag_indexing_allowed`, `rag_pii_exposure_allowed`
Training	`training_permitted`, `training_pii_included`

Full profile (expanded control surface)

The full profile includes the minimal profile plus additional elements from the PII, Extraction, RAG, and Training subsections to express:

- purpose limitation and lawful basis
- component-level scope constraints
- transformation requirements (redaction/masking/pseudonymization)
- caching/index retention and auditability
- model scope restrictions and provenance requirements

Producers SHOULD avoid emitting elements with ambiguous free-text values when a controlled vocabulary is available.

Example

Example use of the governance and compliance elements is shown below:

```
<doclang>
  <head>
    <!-- reserved elements -->
    <title>My Company's Annual Report</title>
    <author_info>
      <author>Author 1 Name</author>
    </author_info>
    <date>2024-01-01</date>
    <language classifier="fastText" score="0.7">eng</language>
    <topic topic_taxonomy="taxonomy" score="0.5">Technology</topic>
    <document_hash
hash_function="sha256sum">75f2db0c6124527bf6dd48440f95fc864a5108d28517633f937923a7d8199
185</document_hash>
    <summary>This is a summary of the document</summary>
    <generated_by>example_vlm_org/example_vlm_name</generated_by>
```

```
    <licenses>
      <license>https://www.apache.org/licenses/LICENSE-2.0</license>
    </licenses>

    <data_classification>
      <data_class>confidential</data_class>
      <data_class>personal information</data_class>
    </data_classification>

    <acceptable_use>
      <purpose>General-purpose language models</purpose>
      <purpose>Sales and marketing</purpose>
    </acceptable_use>

    <stewardship>
      <steward>
        <name>Charles Owens</name>
        <contact>abc@some.org</contact>
        <org>Dataset Organization</org>
      </steward>
    </stewardship>

    <access_policy>
      <policy>
        <ref>http://www.some.org/policies/AC-2345</ref>
        <roles>
          <role>viewer</role>
          <role>reader</role>
        </roles>
      </policy>
    </access_policy>

    <retention_policy>
      <policy>
        <ref>http://www.some.org/policies/AC-2345</ref>
        <retention_period unit="year">5</retention_period>
        <deletion_method>permanent secure deletion</deletion_method>
        <documentation>record deletion event, date, method and personnel
responsible</documentation>
      </policy>
    </retention_policy>

    <compliance_requirements>
      <compliance_req>GDPR</compliance_req>
      <compliance_req>HIPAA</compliance_req>
      <compliance_req>FedRAMP</compliance_req>
      <compliance_req>PCI DSS</compliance_req>
      <compliance_req>EU AI Act</compliance_req>
    </compliance_requirements>

  </head>
  <!-- document content -->
</doclang>
```

Consolidated example (PII + extraction + RAG + training)

The following example illustrates a single `<head>` that combines Privacy and PII controls, Data extraction controls, RAG and retrieval controls, and Document training controls. Implementations MAY choose to interpret these as organization-wide defaults for the DocLang instance, and MAY override at component level for finer-grained control.

```
<doclang>
  <head>
    <!-- core metadata (illustrative) -->
    <title>Customer Support Case File</title>
    <date>2025-01-15</date>
    <language classifier="fastText" score="0.92">eng</language>
    <generated_by>example_pipeline/doc_ingest_v2</generated_by>

    <!-- licensing / classification / compliance (existing governance elements) -->
    <licenses>
      <license>https://www.apache.org/licenses/LICENSE-2.0</license>
    </licenses>

    <data_classification>
      <data_class>confidential</data_class>
      <data_class>personal information</data_class>
    </data_classification>

    <acceptable_use>
      <purpose>Customer support automation</purpose>
      <purpose>Internal analytics</purpose>
    </acceptable_use>

    <compliance_requirements>
      <compliance_req>GDPR</compliance_req>
      <compliance_req>ISO/IEC 27001</compliance_req>
      <compliance_req>ISO/IEC 27701</compliance_req>
    </compliance_requirements>

    <!-- Privacy and PII controls -->
    <pii_status>present</pii_status>
    <pii_sensitivity_level>medium</pii_sensitivity_level>
    <pii_source_type>user_provided</pii_source_type>
    <controller_processor_role>controller</controller_processor_role>
    <pii_processing_purpose>case_resolution</pii_processing_purpose>
    <pii_lawful_basis>legitimate_interest</pii_lawful_basis>
    <pii_minimisation_status>applied</pii_minimisation_status>
    <pii_transformation_level>pseudonymized</pii_transformation_level>
    <reidentification_risk>low</reidentification_risk>
    <access_control_level>restricted</access_control_level>
    <ai_use_restriction>no_general_training</ai_use_restriction>
    <cross_border_transfer_status>not_permitted</cross_border_transfer_status>
    <retention_category>support_case_records</retention_category>
    <dsr_impact_flag>true</dsr_impact_flag>
    <dpia_required>false</dpia_required>
    <logging_monitoring_enabled>true</logging_monitoring_enabled>
```

```
    <!-- Data extraction controls -->
    <extraction_permitted>true</extraction_permitted>
    <extraction_scope>tables_and_forms_only</extraction_scope>
    <extraction_purpose>case_metrics</extraction_purpose>
    <extraction_granularity>field_level</extraction_granularity>
    <pii_extraction_allowed>false</pii_extraction_allowed>
    <sensitive_data_extraction_allowed>false</sensitive_data_extraction_allowed>
    <extraction_transformation_required>redact</extraction_transformation_required>

<extraction_output_constraints>internal_systems_only</extraction_output_constraints>
    <downstream_sharing_permitted>false</downstream_sharing_permitted>
    <extraction_audit_required>true</extraction_audit_required>
    <extraction_audit_retention unit="day">90</extraction_audit_retention>
    <human_in_the_loop_required>true</human_in_the_loop_required>

    <!-- RAG and retrieval controls -->
    <rag_permitted>true</rag_permitted>
    <rag_indexing_allowed>true</rag_indexing_allowed>
    <rag_embedding_scope>summaries_only</rag_embedding_scope>
    <rag_chunking_constraints>max_512_tokens</rag_chunking_constraints>
    <rag_query_restrictions>support_intent_only</rag_query_restrictions>
    <rag_output_attribution_required>true</rag_output_attribution_required>

<rag_output_transformation_required>summarize_and_redact</rag_output_transformation_req
uired>
    <rag_pii_exposure_allowed>false</rag_pii_exposure_allowed>
    <rag_sensitive_data_exposure_allowed>false</rag_sensitive_data_exposure_allowed>
    <rag_downstream_sharing_permitted>false</rag_downstream_sharing_permitted>
    <rag_caching_allowed>true</rag_caching_allowed>
    <rag_cache_retention unit="day">30</rag_cache_retention>
    <rag_audit_required>true</rag_audit_required>
    <rag_audit_retention unit="day">90</rag_audit_retention>
    <rag_model_scope>enterprise_internal_models</rag_model_scope>

    <!-- Document training controls -->
    <training_permitted>false</training_permitted>
    <training_scope>none</training_scope>
    <training_purpose>none</training_purpose>
    <training_model_type>none</training_model_type>
    <training_dataset_reuse_allowed>false</training_dataset_reuse_allowed>

<training_derivative_sharing_permitted>false</training_derivative_sharing_permitted>
    <training_pii_included>false</training_pii_included>
    <training_sensitive_data_included>false</training_sensitive_data_included>
    <training_provenance_required>true</training_provenance_required>
    <training_audit_required>true</training_audit_required>
    <training_audit_retention unit="day">365</training_audit_retention>
    <model_output_usage_constraints>internal_only</model_output_usage_constraints>
    <right_to_be_forgotten_applicability>true</right_to_be_forgotten_applicability>

  </head>
```

```
  <!-- document content -->
</doclang>
```

Notes:

- The example uses illustrative values. Implementations SHOULD define controlled vocabularies for enums such as `pii_status`, `pii_sensitivity_level`, `extraction_scope`, `rag_embedding_scope`, and `training_model_type`.
- When `training_permitted` is `false`, implementations SHOULD treat training-related fields as either omitted or set to explicit "none" values.
- Where retention elements include a `unit` attribute, producers MUST use consistent units and pre-normalized values.